MILITARY SPECIFICATION

SWITCHGEAR, POWER, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope. - This specification covers the requirements for electric power dead front, deck mounted switchgear for shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-387	- Plastic Sheet, Laminated, Thermosetting (for Designation Plates).
QQ-A-200/5	- Aluminum Alloy Bar, Rod, Shapes, and Tube Extruded, 5086.
QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6061 and 6062.
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.
QQ-A-250/8	- Aluminum Alloy, 5052, Plate and Sheet.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-B-613	- Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet, and Strip).
QQ-N-281	- Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, And Structural and Special Shaped Sections.
QQ-S-365	- Silver Plating Electrodeposited; General Requirements For.
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
QQ-S-700	- Steel Sheet and Strip, Medium and High Carbon.
QQ-S-741	- Steel, Carbon: Structural Shapes, Plates, and Bars.
QQ-S-766	- Steel Plates, Sheets and Strip - Corrosion Resisting.
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces For Organic Coatings.
TT-P-645	- Primer, Paint, Zinc-Chromate, Alkyd Type.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

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MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-B-857	- Bolts, Nuts, and Studs.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-D-1000	- Drawings, Engineering and Associated Lists.
MIL-D-1000/2	- Drawings, Engineering and Associated Lists.
MIL-I-1361	- Instrument Auxiliaries, Electrical Measuring: Shunts, Resistors, and Transformers.
MIL-R-2033	- Relays for Naval Shipboard Electrical Service.
MIL-C-2212	- Controllers, Alternating-Current, Naval Shipboard.
MIL-R-2729	- Regulator Sets, Voltage, A.C. Generator (Naval Shipboard Use).
MIL-T-2867	- Thermometers, Indicating, Resistance.
MIL-I-3158	- Insulation Tape, Electrical Glass-Fiber (Resin-Filled); and Cord, Fibrous-Glass.
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-I-7798	- Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic.
MIL-P-8585	- Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
MIL-Q-9858	- Quality Program Requirements.

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MILITATY (cont'd.)

MIL-P-15024 - Plates Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-T-15108 - Transformers, Power, Step-Down, Single Phase, 60-Cycle, 1-KVA Approximate Minimum Rating; and Reactors (Balance Coils)-Dry, Naval Shipboard.
MIL-R-15109 - Resistors and Rheostats, Naval Shipboard.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-F-15160 - Fuses; Instrument, Power, and Telephone.
MIL-S-15291 - Switches, Rotary, Snap Action.
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-I-16103 - Indicator, Phase Sequence.
MIL-I-16104 - Indicators, Synchronization.
MIL-M-16125 - Meters, Electrical Frequency.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-T-16315 - Transformers, Power, Step-Down (Miscellaneous, Naval Shipboard Use).
MIL-E-16366 - Electrical Clamps, Lug Terminals and Conductor Splices - Pressure Grip.
MIL-W-16878/3 - Wire, Electrical, Type D, 105° C., 3000 Volts. (Insulated, High Temperature.)
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Enclosure (Shipboard Use).
MIL-I-17384 - Insulating Compound, Electrical, Quick-Drying.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-17587 - Circuit Breakers, Air, Electric, Open Frame Removable Assembly, (Shipboard Use).
MIL-C-17588 - Circuit Breaker, (Automatic-ALB) and Switch, Toggle (Circuit Breaker, Non-Automatic-NLB), Air, Insulated Enclosure, 125 Volts and Below, AC or DC, Naval Shipboard.
MIL-S-17773 - Switching Unit, Power Transfer (Bus Transfer) (Naval Shipboard Use).
MIL-S-18396 - Switches, Meter and Control, Naval Shipboard.
MIL-W-19088 - Wattmeters, Switchboard Type, 4-1/2 Inch.
MIL-F-19207/1 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL10U.
MIL-F-19207/2 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL11U.
MIL-F-19207/3 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL12U.
MIL-F-19207/21 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL32W.
MIL-F-19207/22 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL33W.
MIL-F-19207/23 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL34W.
MIL-F-19207/24 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL35W.
MIL-R-19523 - Relays, Auxiliary, Naval Shipboard.
MIL-C-21346/1 - Clips, Electrical (Styles FC21 to FC28, Incl.).
MIL-A-22895 - Adhesive, Metal Identification Plate.
MIL-V-23151 - Voltmeter, Expanded Scale, Switchboard Type (Naval Shipboard Use).
MIL-M-23167 - Meter, Frequency, Expanded Scale, Switchboard Type (Naval Shipboard Use).
MIL-S-24188 - Synchronizing Control Equipment, 60 Cycles, 450 Volts, Naval Shipboard.

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings and in Technical-Type Publications.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions Of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.

DRAWINGS

MILITARY

9-S-5302-L - Terminal, Riser, Types AM, BM and BMB for Casualty Power Sym. 1032, 1033 and 1034.

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MILITARY (cont'd.)
9000-S6201-74453 - Switchboard, Appliances, Switchboard Unit, Navy Standard.
9000-S6202-73907 - Light, Indicator (Switchboard) 2 Lamp, SPF, Types B-27A thru B-27G.
9000-S6202-74307 - Indicator Light, Switchboard, 2-Lamp for DC Circuit 2628.1 and 2628.2.

PUBLICATION

NAVSHIPS
NAVSHIPS 362-2352 - Wire Bundling Rating Manual.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A-131 - Structural Steel For Ships.
A-415 - Hot Rolled Carbon Steel Sheets.
B-187 - Copper Bus Bar, Rod, and Shapes.
B-317 - Aluminum-Alloy Extruded Bar, Rod, Pipe, and Structural Shapes for Electrical Purposes (Bus Conductors).

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Definitions of terms. - The following definitions shall apply to the terms listed below wherever such terms appear in this specification:

3.1.1 Standardized unit. - A standardized switchgear unit is a vertical subdivision of a section complete with separate front panels and all required appurtenances constructed in accordance with Drawing 9000-S6201-74453.

3.1.2 Section. - A switchgear section is a self-contained assembly of switchgear units consisting of a complete framework with all required enclosures and with a common base mounting.

3.1.3 Switchgear group. - A switchgear group is synonymous with the term switchboard; however, switchboard groups may consist of two or more individual sections rather than all units being assembled together into a single structure.

3.1.4 Control benchboard. - A control benchboard is a switchgear section consisting of a switchboard and a control desk combined in a single structure with the top of the control desk, either horizontal or inclined, attached to the front switchboard panels. It is provided with the necessary instruments and control and indicating apparatus for the centralized control of generator, bus tie, and feeder circuits (also to include the battery circuits on submarines) of one or more switchgear groups.

3.1.5 Vertical control switchboard. - A vertical control switchboard is similar to a control benchboard except that all equipment is mounted on vertical panels.

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3.1.6 Load center. - A load center distribution section is a section consisting of one or more distribution units. It serves as a power distribution center in locations requiring power supplies exceeding the capacity of distribution panels.

3.1.7 Nonstandard switchgear. - Nonstandard switchgear is switchgear consisting of units, sections, and groups, not conforming to the standardization requirements of Drawing 9000-S6201-74453.

3.2 General requirements. - The switchboards shall be in accordance with the following general requirements of MIL-E-917. When a requirement of MIL-E-917 conflicts with a requirement of this specification, the requirement of this specification shall govern:

(a) Materials.
(b) Metals or coating.
(c) Electrical insulation.
(d) Soldering.
(e) Brazing.
(f) Threaded parts.
(g) Inclination.

3.2.1 Whether or not specifically mentioned for the various units, all switchgear groups shall be furnished complete with all appurtenances required to accomplish the function specified hereinafter.

3.2.2 No mercury shall be used in any process of manufacture or test of switchgear and its components.

3.3 Ship service power switchgear group for ships having alternating current (a.c.) power. -

3.3.1 A.C. generator control unit for switchgear groups having a control benchboard (see figure 1 for a typical one line diagram). - The following appurtenances shall be mounted on or within the unit:

(a) One generator circuit breaker.
(b) One control power transformer 450/120 volt (V)., connected to generator side of circuit breaker, for generator circuit breaker and governor motor control circuits, if required.
(c) Potential transformers, for metering, relaying, electric governor, and synchronizing, in number as required.
(d) Current transformers, for metering and relaying, electric governor, in number as required. Provide protective device when metering is remote.
(e) One current transformer, complete with protective device for remote metering, where remote metering is required.
(f) Fuse panel for fusing of the potential and control transformer primary circuits and any control circuits extending outside of the section.
(g) Space, mounting and wiring, for the voltage regulator equipment, including associated current and potential transformers. All equipment shall be furnished with the generators.
(h) 1/ One circuit breaker (see 3.15), type AQB-A or AQB-LF, 500 V., 3-pole, 250-ampere (amp.) frame size. The circuit breaker shall be connected to the generator bus ahead of the generator circuit breaker.
(i) 1/ One casualty power terminal, riser type, conforming to Drawing 9-S-5302-L connected through the AQB-A or AQB-LF circuit breaker, item (h).
(j) One reverse power relay, single phase, to prevent the generator from motoring on the bus after loss of driving power from prime mover.
(k) One indicator light (blue) to show the closed position of the generator circuit breaker.
(l) One generator running and voltage available auxiliary relay for control of associated indicator lights.

3.3.2 A.C. generator control unit for switchgear groups not having a control benchboard (see figure 2 for a typical one-line diagram). - Appurtenances listed in 3.3.1, except item (l), shall be mounted on or within the unit.

3.3.2.1 The following devices shall be mounted on the front panel of the generator unit:

(a) One a.c. ammeter, for reading one phase of generator current.
(b) One a.c. voltmeter with transfer switch, for reading one phase of generator potential and one phase of each bus tie voltage and one phase of shore power voltage for switchgear group containing shore power connection.

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(c) One polyphase wattmeter, for reading generator output.
(d) One frequency meter for reading the frequency of the generator.
(e) One synchroscope and one set of synchronizing lights with transfer switch for synchronizing the generator with the main bus, and the main bus with the bus tie circuit and the bus tie with shore power for switchgear group containing shore power connection.
(f) One control switch for control of the generator circuit breaker where an electrically operated breaker is used.
(g) Control switches, rheostats, and similar devices, as required in conjunction with the governor speed regulation of the generator prime mover.
(h) One transfer switch to transfer the governor control from the local control switch to the remote control switch on the a.c. bus tie unit (or control unit) of the control switchgear group. It shall also control the lamp (see 3.3.4.1(c)), on the bus tie unit or control unit to indicate when the remote switch is in control. This shall be provided only when the unit has control of a remote generator.
(i) One control switch with white indicator light for control of the generator space heaters when provided with the generator.
(j) Control switches, rheostats, and similar equipment that is designed for panel mounting and is required in conjunction with the voltage regulator equipment mounted in the generator unit shall be mounted and operated from the front panel.
(k) For installations in which a bus tie unit is not provided, one set of three indicator lights with a control switch (see 3.3.4.1(i)) shall be mounted on the front panel of the generator unit.
(l) 2/ One synchronizing control monitor with required accessories, as specified in MIL-S-24188, shall be provided for each single or two generator switchgear group. The monitor shall be connected to electrically prevent the closing of the generator circuit breakers, bus tie circuit breakers, and shore power circuit breaker as applicable, unless the systems to be paralleled are properly synchronized.

1/ When the generator capacity is in excess of 400 kilowatts (kw) the quantity of 3.3.1(h) and (i) shall be furnished as specified (see 6.1).
2/ The synchronizing control monitor may be mounted in the bus tie unit instead of the generator control unit.

3.3.3 A.c. bus tie unit for switchgear groups having a control benchboard. - The following appurtenances shall be mounted within the unit:

(a) One circuit breaker.
(b) One feeder circuit breaker.
(c) Necessary disconnecting devices for performing switching functions as may be required.
(d) Potential transformers, for metering, relaying, and synchronizing, in number as required for the bus connections of the switchgear group.
(e) One current transformer for metering the current interchange through the bus tie circuit breaker.
(f) Fuse panel for fusing the potential and control power transformer primary circuits.
(g) Indicator light (blue) to show the closed position of the bus tie and feeder circuit breaker.

One bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus-tie and shore power unit". In addition this unit shall substitute item (h) for the feeder circuit breaker, and add item (i) through (k) below:

(h) 1/ One shore power connection circuit breaker. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. The current transformer (see 3.3.3(e)), shall be located so that it is in the circuit when feeding power to the local switchgear group. Where a 3200 amp. frame size circuit breaker is required, a separate unit may be provided.
(i) One indicator light (blue) to show the closed position of the shore power circuit breaker.
(j) One current transformer for metering shore power current.
(k) Potential transformers for metering and synchronizing, in number as required.

1/ To be furnished on only one bus tie unit of any switchgear group.

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3.3.4 A.c. bus tie unit for switchgear groups not having a control benchboard. - The following appurtenances shall be mounted within the unit:

(a) One circuit breaker.
(b) One feeder circuit breaker.
(c) Necessary disconnecting devices for performing switching functions as may be required.
(d) Potential transformers, for metering, relaying and synchronizing, in number as required for the bus connections of the switchgear group.
(e) Fuse panel for fusing the control circuits fed from the control power bus.
(f) Fuse panel for fusing the potential and control power transformer primary circuits.

3.3.4.1 The following devices shall be mounted on the front panel of the bus tie unit:

(a) One single element wattmeter, calibrated to indicate the three phase power output of each remote ship's service generator.
(b) One a.c. ammeter for indicating the current of each remote ship's service generator and from the shore power connection.
(c) Two remote control switches or rheostats, and indicator lights (white), as required, to control the speed of the prime movers of the remote generators. The control switch power shall be connected through the transfer switches (see 3.3.2.1(h)) on the generator units of the remote switchgear groups, and the indicating lamps shall be connected to indicate when the transfer of control is accomplished. This shall be provided when the unit has control of a remote generator.
(d) One composite transfer switch for reading the power and current of each remote generator by means of (a) and (b) above and the shore power current by means of (b) above.
(e) One control switch for control of the bus tie circuit breaker where an electrically operated circuit breaker is used.
(f) Indicator lights (blue) to show the closed position of the bus tie circuit breaker and the type ACB (see 3.15) feeder circuit breaker.
(g) One frequency meter for reading the frequency of the bus tie circuit.
(h) One transfer switch, with two white indicator lights, for connecting the switchboard control bus to the local 450 V. ship service bus, or the 450 V. bus of the corresponding (forward or aft) emergency switchboard. Transfer may be accomplished automatically; however, in such cases one source of power shall be the local 450 V. ship service bus and the second source shall be a 450 V. bus tie supply.
(i) One set of three indicator lights with a control switch to indicate ground conditions on the main bus or any of its connected feeders.

3.3.4.2 One bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus tie and shore power unit". In addition to the equipment specified in 3.3.4 and 3.3.4.1, this unit shall substitute item (a) for the feeder circuit breaker, and the additional items (b) through (g) shall be provided:

(a) One shore power connection circuit breaker. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker.
(b) One indicator light (blue) to show the closed position of the shore power circuit breaker.
(c) Two indicator lights (white), one to indicate the shore power feeder energized and the other to indicate the bus tie feeder energized. These two indicator lights shall be mounted near the shore power circuit breaker.
(d) One current transformer for metering the shore power current.
(e) One switchboard type phase sequence indicator conforming to type II of MIL-I-16103 with a control switch for use in checking phase rotation of shore power connections.
(f) Potential transformers for metering and synchronizing, in number as required.
(g) One transfer switch for use with item (g) of 3.3.4.1 for reading the frequency of the bus tie circuit and shore power.

3.3.5 Control benchboard or vertical control switchboard. - The control benchboard shall be or the bench type with vertical panels for the mounting of instruments and control equipment, a horizontal of inclined desk top and an apron section below. Control benchboards for surface ships shall be provided with a chair. It shall be so mounted that it can be swung out of the way when not in use. Control switches for the electrically operated circuit breakers controlled from the benchboard shall be mounted on the desk top and arranged within convenient reach of the operator. The circuit breaker control switches and their corresponding switches in

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the mimic bus system shall be located so that their relative positions as to their location on the benchboard are similar. The manual control rheostats shall be located in the apron portion so as to provide a maximum of free panel space for mounting of equipment. The portion of the apron panel used for mounting of the equipment shall not project beyond the front edge of the desk top. Space shall be provided for filing the framed data sheets. Vertical control switchboard shall be furnished when specified (see 6.1).

3.3.5.1 Surface ship control benchboard. - The following devices shall be mounted on the control benchboard for surface ships:

(a) Control switches, rheostats and similar equipment that is designed for panel mounting and is required in conjunction with the voltage regulator equipment mounted in the generator unit, shall be mounted on and operated from the control benchboard.
(b) One a. c. ammeter for each local generator for reading one phase of generator current.
(c) One a. c. voltmeter for each local generator for reading one phase of generator potential.
(d) One a. c. voltmeter with transfer switch for reading single phase voltage of the switchboard bus and of the bus tie feeders and shore power for switchgear groups containing shore power connection.
(e) One polyphase wattmeter for reading generator output for each local generator.
(f) One single element wattmeter with transfer switch calibrated to indicate the 3 phase power output of the remote ship service generator. (Shall be provided where a switchgear group is to have control of a remote generator or generators.)
(g) One temperature indicator complete with selector switch and accessories when temperature detectors are provided with the generator.
(h) 1/ One frequency meter for reading the frequency of the generator for each local generator.
(i) 1/ One frequency meter with transfer switch for reading the frequency of the bus and of each bus tie circuit and shore power for switchgear group containing shore power connection.
(j) One synchroscope and one set of synchronizing lights with transfer switch for synchronizing the generator with the bus and the bus with the bus tie circuits and the bus tie with shore power for switchgear group containing shore power connection. Where the switchgear group has control of two generators and two bus tie circuits the synchroscope and synchronizing lights shall be connected for synchronizing either generator with the bus and the bus with either bus tie circuit.

1/ For units controlling two local generators, each frequency meter, item (h), shall be provided with a transfer switch for reading the frequency of one generator and one bus tie circuit and shore power, in which case item (i) shall be omitted.

(k) One local control switch or other control device as required, for control of the speed governor of the generator prime mover for each local generator.
(l) One transfer switch to transfer the governor control of the generator prime mover from local control to the remote control on the master control benchboard. It shall also control the indicator light on the master control unit to indicate when the remote station is in control. This transfer switch shall be provided only on control benchboards for which remote control, item (m), is to be provided on the master control benchboard.
(m) One remote control switch, or other remote control devices as required, and indicator light (white) to control the speed governor on the remote generator prime mover where the unit has control of a remote generator. The control power shall be connected through the transfer switch, item (l), of the remote control benchboard, and the indicating light shall be energized when the transfer of control to the remote station is accomplished.
(n) One control switch with indicator light (blue) for control of each local generator circuit breaker.
(o) One control switch with indicator light (blue) for control of each bus tie circuit breaker.
(p) One control switch with indicator light (blue) for control of each electrically operated distribution circuit breaker.
(q) One control switch with white indicator light for control of generator space heater (when provided with the generator) for each local generator.
(r) Test receptacles with test plugs for connecting portable instruments to the metering current and potential circuits.
(s) One transfer switch with two white indicator lights for connecting the switchboard control bus to the local 450 V. ship service bus, or the 450 V. emergency supply. The transfer may be accomplished automatically however, in such cases one source of power shall be the local 450 V. ship service bus and the second source shall be a 450 V. bus tie supply.
(t) One set of three indicator lights with a control switch to indicate ground conditions on the main bus or any of its connected feeders.

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(u) Fuse panel for fusing the control circuits.
(v) One mimic bus system (dark type) simulating the generator and bus tie connections and shall consist of the following: Indication lights (white) for each generator to indicate generator running and voltage available. Indication lights (amber) for each generator and bus tie circuit breaker. Single pole, double throw switches shall be provided for each circuit breaker light. These switches shall be combined with their respective amber lights and so arranged that when the switch handle indication and the circuit breaker position (open or closed) are the same, the light shall be out. When the circuit breaker position and the switch handle indication are different, the light shall be lit. For indicator light power supply, (see 3.14.7). Mimic bus material shall be 1/8 inch thick, 1/4 inch wide and painted black. Mimic bus material shall be laminated plastic conforming to type NDP or GCP-H of L-P-387, aluminum alloy conforming to QQ-A-250/8, commercial brass conforming to QQ-B-613, corrosion resisting steel conforming to class 316 of QQ-S-766 or nickel-copper-alloy (monel) conforming to QQ-N-281.
(w) One ammeter for reading interchange of current in the bus tie.
(x) One ammeter for each remote generator for reading one phase of generator current.
(y) One shore power circuit breaker control switch with one blue indicator light to indicate breaker closed. Two white indicator lights, one to indicate shore power feeder energized and the other to indicate the bus tie feeder energized. These two indicator lights shall be mounted near the shore power circuit breaker control switch.
(z) One control switch with one blue and one white indicator light for each electrically operated circuit breaker feeding an emergency switchboard.
(aa) One ammeter for reading the shore power current.
(bb) One switchboard type phase sequence indicator conforming to type II of MIL-I-16103 with a control switch, for use in checking phase rotation of shore power connections.
(cc) One synchronizing control unit with required accessories as specified in MIL-S-24188, shall be provided for each single or each two generator switchgear group. The unit shall be connected to electrically prevent the closing of the generator circuit breakers, bus tie circuit breakers, or shore power circuit breakers as applicable, unless the systems to be paralleled are properly synchronized.

3.3.5.2 Submarine control benchboard. - The following devices shall be mounted on submarine control benchboards:

(a) For each ship service turbine generator: items (a), (b), (e), (k), (n) and (q) and (aa) of 3.3.5.1.
(b) For each ship service motor generator: items (a), (b), (n) and (q) of 3.3.5.1 and the following additional devices:
  (1) Control switches, rheostats, and similar devices required in conjunction with speed regulating equipment.
  (2) One control switch with indicator light (blue) for control of each motor circuit breaker.
  (3) One d.c. ammeter, zero center scale for reading d.c. current.
(c) For a diesel-generator.
  (1) Control switches, rheostats and similar equipment that is designed for panel mounting and is required in conjunction with the voltage regulator equipment mounted in the generator unit, shall be mounted on and operated from the control benchboard.
  (2) One control switch with indicator light (blue) for control of the diesel-generator circuit breaker.
  (3) One a.c. ammeter for reading one phase of the generator current.
  (4) One polyphase wattmeter for reading the generator power output.
  (5) One control switch or other control devices as required, for control of the governor on the generator prime mover.
(d) For the battery:
  (1) Two zero center scale d.c. ammeters, one high range and one low range, to measure battery charge and discharge currents.
  (2) One battery float charge measuring device.
  (3) One hydrogen detector remote indicator.
  (4) One control switch with indicator light (blue) for control of each battery circuit breaker.
  (5) One battery exhaust fan control switch with indicator light (green).
  (6) One battery ventilation flow indicator.
  (7) One control switch with ammeter to control and indicate operation of battery recirculating fans.
  (8) One ampere-hour meter (non-mercury type) with associated equipment.

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(e) For the system:
(1) One synchroscope with synchronizing selector switch for synchronizing generators with the bus and the bus with bus tie circuits.
(2) Two frequency meters and two voltmeters connected through the synchronizing selector switch for reading frequency and voltage on each side of the point at which synchronizing is being performed.
(3) One d. c. voltmeter with selector switch for reading battery and d. c. bus voltages.
(4) One transfer switch with two indicator lights (white) for connecting the a. c. control bus to a 450 V. normal and alternate source of supply.
(5) One transfer switch with two indicator lights (white) for connecting the d. c. control bus to a normal and alternate d. c. supply.
(6) A direct reading ground detector.
(7) Fuse panel for fusing control circuit.
(8) One mimic bus system simulating the generators, bus ties, shore power and battery connections of the power system (see 3. 3. 5. 1, item (v)).
(9) A transfer device with indicator light (white) shall be installed for supplying each d. c. control bus from either its respective motor-generator (MG) d. c. bus or from the battery side of the battery circuit breaker.

3. 3. 6 ACB circuit breaker distribution unit. - This unit shall provide for the distribution of power from the main bus. The following appurtenances shall be provided:

(a) Circuit breakers, having current ratings and protective features as required.
(b) Indicator lights (blue) mounted on the front panel of the unit or on the circuit breaker escutcheon to indicate when each of the feeder circuit breakers is closed.
(c) Control switches shall be provided for electrically operated circuit breakers. Where the switchgear group has a control benchboard, these switches shall be located on the control benchboard.
(d) Necessary disconnecting devices for disconnecting cables between switchgear sections, and performing other switching functions.

Note. - Where a circuit breaker supplies a feeder which may be energized from another source such as the emergency switchgear group, a white light shall be provided to indicate feeder energized.

3. 3. 7 AQB circuit breaker distribution unit (see figure 3). - This unit shall provide for the distribution of electric power or lighting supply (450 or 120 V) through type AQB, 500 V., 3-pole circuit breakers. Units receiving two sources of supply shall be provided with interlocked NQB or AQB circuit breakers. Where interlocked circuit breakers are installed, an indicator light shall be provided for each power supply to indicate when power is available (see 3. 6).

3. 3. 7. 1 Spaces for the future installation of circuit breakers resulting from the use of standardized distribution units shall be blanked off, with busing installed for the circuit breakers. Mounting blocks shall also be installed for AQB circuit breakers.

3. 3. 8 ACB - AQB circuit breaker distribution unit. - This unit shall provide for distribution of power through both ACB and AQB type circuit breakers. It shall contain the appurtenances listed in 3. 3. 6 through 3. 3. 7. 1.

3. 4 Emergency power switchgear groups for vessels having a. c. power. - Switchgear groups for controlling larger than 100 kw. generators shall employ a three-way type ABT-A3 (ES) circuit breaker type bus transfer equipment, and shall be constructed as sectionalized switchgear groups having a separate generator and bus transfer section, and feeder sections. For switchboards controlling emergency generators rated 100 kw. and less, the bus transfer equipment, type ABT-A3 (ES), contactor type shall be used. When the contactor type unit is used, the switchboard shall be assembled with the AQB distribution units to form a single section (see figures 4 and 5). Bus transfer equipment shall be in accordance with MIL-S-17773.

3. 4. 1 A. c. emergency generator control and circuit breaker type bus transfer section. - The following appurtenances shall be mounted on or within the unit:

(a) One type ABT-A3 (ES), circuit breaker type, 3-way, automatic bus transfer equipment having a current rating not less than the continuous rating of the emergency generator and the current rating of the feeders from the ship service switchboards, as specified. The associated control

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switches, indicator lights, and relays suitable for panel mounting shall be mounted on one of the front panels.

(b) Necessary disconnecting devices for disconnecting cables between switchgear sections, and performing other switching functions.

(c) Two potential transformers for metering and relaying.

(d) One type AQB-A circuit breaker 100 amp. frame size, 500 V., 3-pole, instantaneous element only, for power supply to the booster pump supplying cooling (salt) water to the diesel engine as specified (see 6.1). This circuit breaker shall be connected to the generator bus ahead of the generator circuit breaker. A red warning instruction plate shall be provided stating that the circuit breaker shall always be in the closed position when the generator is set-up for starting.

(e) Two current transformers for metering and relaying.

(f) One fuse panel for fusing the potential transformer primary circuits.

(g) 1/ One circuit breaker type AQB-A or AQB-LF, 500 V., 3-pole, 250 amp. frame size. The circuit breaker shall be connected to the generator bus ahead of the generator circuit breaker.

(h) 1/ One casualty power terminal, riser type, Drawing 9-S-5302-L, connected through the AQB-A or AQB-LF circuit breaker, item (g).

1/ When the generator capacity is in excess of 400 kw the quantity of 3.4.1(g) and (h) shall be furnished as specified (see 6.1).

3.4.1.1 The following devices shall be mounted on the front of the panel:

(a) One a. c. ammeter for reading one phase of generator current.

(b) One a. c. voltmeter for reading one phase of generator potential.

(c) One polyphase wattmeter for reading generator output.

(d) One frequency meter for reading generator frequency.

(e) One control switch with white indicator light for control of the generator space heaters when provided with the generator.

(f) Instruction plates containing complete operating instructions for the automatic equipment and a single line diagram of the main buses.

(g) One set of three indicator lights and switch for ground indication on the 450 V. emergency power bus or any of its connected feeders.

(h) One indicator light with green color cap and so connected as to indicate when all manually operated control devices are setup for automatic operation. The applicable following functions and conditions shall be included to complete the green light circuit:

(1) Generator circuit breaker (closed) (when separated from bus transfer).
(2) Voltage regulator transfer switch (automatic position).
(3) Ship service bus transfer equipment (set for automatic operation).
(4) Emergency bus transfer equipment (set for automatic operation).
(5) Diesel starting control circuits (set for automatic starting).
(6) Diesel starting air shut-off valve (open).
(7) Electrically operated emergency bus unloader switch (if used) (set for automatic operation).

(i) Space mounting and wiring for the voltage regulator equipment, including associated current potential transformers, and for the exciter field rheostat, all equipment to be furnished with the emergency generator. The control switches, rheostats, and similar equipment designed for panel mounting shall be suitably located for front-of-panel operation.

(j) Space mounting and wiring for equipment, for the control switches and interlocks, furnished with the bus transfer equipment. The control switches and similar equipment designed for panel mounting shall be suitably located for front-of-panel operation.

3.4.2 A. c. emergency generator control and contactor type bus transfer section. - This unit shall be equivalent functionally to the section specified in 3.4.1. The appurtenances shall be the same as required in 3.4.1, except that the following shall be provided in lieu of 3.4.1(a):

(a) One generator circuit breaker, type AQB-A, 250 amp, frame size, 500 V., 3 pole, manually operated.

(b) Two type ABT-A3 (ES), contactor type bus transfer equipments having current ratings not less than the current rating of the feeders from the ship service switchboards, as specified. One equipment shall be connected for selection of the ship's service supply, the second equipment shall be connected for transfer between ship service supply and the emergency generator supply.

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(c) Four indicator lights (blue) to indicate closed positions of the contactors of the type ABT-A3 bus transfer equipments. The indicator light circuits shall be wired through auxiliary contacts provided in the bus transfer equipments.
(d) Two indicator lights (white) to indicate that the ship's service power feeders are energized.
(e) One type NQB 500-V, 3-pole circuit breaker, connected to by-pass the bus transfer equipment selecting between ship service and emergency so that emergency generator power may be supplied to the ship service switchboards when their generators are secured.
(f) Two control switches, one for each bus transfer equipment, for selecting between manual and automatic control of the equipments.
(g) One electrically operated emergency bus unloader switch complete with control relay and current transformer. This switch shall be connected between the feeder for the steering power switchboard and the emergency bus to disconnect the emergency bus during periods of high current requirements by the steering gear auxiliaries when energized from the emergency generator. The equipment shall function to hold the unloader switch closed unless the demand on the steering power switchboard exceeds 150 percent of the full load current of two steering motors plus 100 percent of the full load current of all other steering gear auxiliaries, in which case the unloader switch shall open provided the power is being supplied by the emergency generator. This item shall be furnished only when specified (see 6.1).

3.5 Distribution units. - The distribution units of the emergency switchgear group shall be in accordance with 3.3.6 through 3.3.8.

3.6 Load center distribution sections. - These sections shall be switchgear unit type construction. The units comprising the section shall provide for distribution through type ACB or AQB circuit breakers, and shall be constructed in accordance with 3.3.6 through 3.3.8.

3.6.1 Distribution units receiving two sources of supply shall be provided with interlocked NQB or ACB circuit breakers. Where interlocked circuit breakers are installed, an indicator light shall be provided for each power supply to indicate when power is available.

3.6.2 Each load center distribution section shall be provided with two type AQB-A or AQB-LF 500 V., 3-pole, 250 amp. frame size circuit breakers for casualty power supply and two casualty power terminals, riser type conforming to Drawing 9-S-5302-L, connected through the type AQB circuit breakers to the main bus.

3.7 Ship service power switchgear group for d.c. vessels. -

3.7.1 D.C. generator control unit. - The generator control unit shall contain the following appurtenances:

(a) One circuit breaker.
(b) One d.c. voltmeter and switch for reading bus and generator voltage.
(c) One d.c. ammeter complete with shunt for reading generator current.
(d) One indicator light (blue) to show the closed position of the generator circuit breaker.
(e) One set of two ground detector lamps and one control switch for indication of ground conditions on d.c. bus or its connected feeders.
(f) Space, mounting and wiring, for the generator field rheostat and operating mechanism to be furnished with the generator.
(g) One reverse current relay to prevent the generator from motoring on the bus after loss of driving power from prime mover. (Provide only where more than one generator is connected to the d.c. bus.)

Note. - The lower section of the unit may be used for the mounting of distribution switches or circuit breakers where the utilization of this space will result in a saving in the overall size of the switchboard.

3.7.2 D.C. bus tie unit. - The following appurtenances shall be mounted within the unit:

(a) One circuit breaker.
(b) Necessary disconnecting devices for performing switching functions as may be required.
(c) Fuse panel for fusing control circuits.
(d) Shunt for bus tie ammeter (in positive bus only).

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3.7.2.1 The following devices shall be mounted on the front panel of the bus tie unit:

(a) One indicator light (blue) to indicate the closed position of the bus tie circuit breaker.
(b) One ammeter with zero center scale for indicating power flow in the bus tie. The ammeter shall be connected to read to the right for power flowing from the bus to the bus tie.
(c) Voltmeter with transfer switch for reading the voltage of the switchboard bus and of the bus tie.

Note. - The lower section of the unit may be used for the mounting of distribution switches or circuit breakers where the utilization of this space will result in a saving in the overall size of the switchboard.

3.7.2.2 The bus tie unit of the switchgear group containing the shore power connection shall be designated as a "bus tie and shore power unit". In addition to the equipment specified in 3.7.2 and 3.7.2.1, this unit shall have the following:

(a) One shore power connection circuit breaker. The shore power circuit shall be connected to the bus tie circuit so that power may be fed to either the local or remote switchboard through a bus tie circuit breaker. The ammeter shunt (see 3.7.2(d)), shall be so located that it is in the circuit when feeding power to the local switchboard.
(b) One indicator light (blue) to indicate the closed position of the shore connection circuit breaker.
(c) One indicator light (white) to indicate when the shore connection feeder is energized.
(d) One indicator light (white) to indicate when the bus tie feeder is energized.
(e) Warning plate adjacent to the indicator lights for the shore connection and bus tie feeder stating "Do not close this circuit breaker when the bus tie circuit is energized.

3.7.2.3 For installations not requiring a bus tie unit, items (a), (b) and (c) of 3.7.2.2, shall be mounted on one of the ACB distribution circuit breaker units and the shore power circuit connected to the switchboard bus. In such instances, this unit shall be designated as "ACB distribution circuit breaker and shore power unit", and the following additional equipment shall be provided:

(a) One indicator light (white) to indicate when the switchboard bus is energized.
(b) Warning plate adjacent to the indicator lights to warn against closing the shore connection circuit breaker when the switchboard bus is energized.

3.7.3 D.c., AQB circuit breaker distribution unit. - This unit shall provide for the distribution of power or lighting through type AQB, 250 V., d.c. circuit breakers.

3.8 Switchboard structure and its compartmentation. -

3.8.1 General construction. -

3.8.1.1 Standardized switchgear sections. - Each unit of a switchgear section shall be in accordance with the applicable unit type shown on Drawing 9000-S6201-74453 and applicable notes on this drawing. The units shall be constructed with a self-supporting, boxlike framework for support of the front panels, busswork, circuit breakers, and other appurtenances to be included in the various units which constitute the section. The number and size of circuit breakers and overall dimensions of the unit shall be in accordance with Drawing 9000-S6201-74453. Units shall be bolted together to form sections and shall be connected by transverse buses. Bus locations shall be as shown on Drawing 9000-S6201-74453, except that where bus locations are not so specified, they shall be installed to suit the application. Each unit and enclosure shall be constructed to meet the mechanical shock requirements of MIL-S-901 when tested as specified in 4.3.4.

3.8.1.2 Special switchgear sections. - Each section of a switchgear group shall be constructed with a rigid, self-supporting, boxlike framework for support of the front panels, busswork, circuit breakers, and other appurtenances to be included in the various units which constitute the section. The framework for a section may be constructed as an integral frame for all units, using common vertical and horizontal members at the junction of adjacent units, or may be formed by constructing an independent box framework for each unit joining the individual units by bolting or welding to form the section. Where each edge of the unit is formed by welding formed members, the ends joining to form a section shall be welded including necessary intermediate welding of the joining members to add strength and rigidity to the section. Each unit and enclosure shall be constructed to meet the mechanical shock requirements of MIL-S-901 when tested as specified in 4.3.4.

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3.8.2 Dimensions. - Insofar as practicable, the grouping of the various units which comprise a switchgear group shall be such that the length of each section is approximately equal to the height; however, the length of a section shall not exceed approximately 8 feet. In the arrangement of the appurtenances to be located within the structure, and the general design of the unit, particular attention shall be given to keeping the center of gravity of the unit as low as practicable.

3.8.3 Framework fabrication. -

3.8.3.1 Standardized and special units. - The framework shall be fabricated from angles, channels, or other structural shapes, or formed members. Each edge of each unit shall be formed by a length of a structural shape, formed member or formed members welded to effect a continuous length. Intermediate vertical supporting members shall be provided for the support of such heavy appurtenances as circuit breakers and large disconnecting switches. Gusset plates shall be welded at the corners of the units in both lengthwise and lateral vertical planes such that each corner shall have a strength not less than the bending strength of the weakest adjoining vertical or horizontal member. Typical bottom corner construction is shown on figure 6.

3.8.4 Gusset plates. - In the special case where the lengthwise gusset plates at the front of the unit interfere with appurtenances, special consideration shall be given to the design of the corner to provide maximum strength. Small gusset plates or L-shaped plates may be used in the above special case. Gusset plates shall not be less than 1/8 inch thick; however, where the thickness of the gusset is less than 3/16 inch, the diagonal edge shall be bent to form a flange. Three sided, forged or formed and heat treated aluminum alloy gussets may be used on aluminum constructed switchboards. Typical gusset plates is shown on figure 7.

3.8.5 Framework assembly. - The framework of each standardized unit or special section shall be stiffened to resist bending by the use of diagonal braces or welded sheets, but braces and sheets shall not be secured in such manner as to restrict access to the front and rear of the units. The framework of each unit or section shall be fabricated by welding all members except those which must be disassembled for replacement of appurtenances. In these cases bolting shall be employed. Riveting shall not be used in the assembly. Interframe bolting of standardized units to form sections shall be in locations as shown on Drawing 9000-S6201-74453. The minimum size of bolt used for this purpose shall be 1/2 inch diameter. The structural shapes that comprise the base members shall not be less than 1/4 inch thick and the horizontal flanges which are bolted to the foundation shall not be less than 2 inches wide. The members in the top of the frame of all switchboards shall have sufficient strength and rigidity for securing top braces. The braces shall not be considered as part of the switchboard.

3.8.6 Bolting to foundation channel. -

3.8.6.1 Standardized units. - Bolting to foundation channel shall be as shown on Drawing 9000-S6201-74453. Where the thickness of the frame member is less than 3/8 inch, a 1/4 inch thick reinforcing washer, approximately 2 inches square, shall be provided under each bolt head.

3.8.6.2 Special sections. - The frame of each section shall be bolted directly to the channel by 3/4 inch diameter bolts conforming to MIL-B-857. Bolt holes in channels shall be 25/32 inch. The minimum number of holding-down bolts to be provided for a section shall be based on the total weight of the section, allowing 210 pounds for each 3/4 inch diameter bolt. Not less than 4 bolts shall be provided per unit. The holding-down bolts shall be located as near the vertical frame member as practicable, but with not more than 3 inches horizontal separation between the vertical member and the center of the bolt hole. Where more than 12 bolts are required, two in each corner, and two adjacent in each intermediate vertical member, the additional bolts shall be located as near the corner bolts as practicable. Where the thickness of the frame member is less than 3/8 inch a 1/4 inch thick reinforcing washer, approximately 2 inches square, shall be provided under each bolt head.

3.8.7 Side sheets. - The ends of each standardized unit or special section shall be enclosed by solid sheets, at least 1/16 inch thick, welded to the switchboard framework, except where the arrangement of switchgear within the section is such as to require side access for maintenance, in which case the side sheet or a portion thereof shall be removable. Removable side or rear sheets are not considered to be structural members and the tests required shall be made with such sheets removed. The side sheets shall be welded by intermittent welds or spot welds to all vertical and horizontal frame members. The gusset plates in the plane of the sheet may be omitted, provided the sheet is welded to the frame at the corners with the same length of welding that would be normally employed with a gusset plate. When gusset plates are omitted, plug welding of the side sheets at the top corners of the frame only is permitted. Side sheets shall be cut to provide for

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interconnecting bus bars and wiring, and suitable insulating spacers shall be employed to close the openings. Location of cutouts for busbars shall be as shown on Drawing 9000-S6201-74453.

3.8.8 Compartmentation. - Each unit shall be internally subdivided by sheet metal barriers to form separate compartments for the circuit breakers to isolate them from the bus work and cable connections. When barriers are installed insulation distances shall be as specified in 3.12. A separate compartment shall be provided for each type ACB circuit breaker, but all type AQB circuit breakers on one unit may be in a common compartment. The compartments shall be constructed sufficiently tight to prevent an arc or fault from spreading from one compartment to another or into the area occupied by the bus work. Bus disconnecting devices shall be located outside the circuit breaker compartment. When the rear enclosure of the AQB circuit breaker compartment is used for support of the circuit breakers, the terminal mounting block shall be bolted through reinforcing angles or channels secured to the sheet enclosure. Each type ACB circuit breaker compartment shall be enclosed on the front by a separate hinged or removable panel. The type AQB circuit breaker compartments shall be enclosed on the front by a removable panel section. On type AQB circuit breaker compartments over 60 inches high, the front enclosure shall be made by two panels.

3.8.9 Ventilation. - The compartments shall be provided with adequate ventilation for cooling of equipment. In providing means for ventilation, consideration shall be given the fact that the switchboards may be operated under conditions favorable to condensation of water vapor within the unit. Type ACB circuit breakers shall be vented through the front of the unit or through the rear of the breaker compartment into the bus bar compartment. Ventilation for type AQB circuit breaker compartments shall be provided by allowing air to enter the compartment below the circuit breaker mounting sheets and escaping from the compartment through an opening above the top row of circuit breakers. The area of the openings at the top and the bottom of the compartment shall be based on allowing not less than 6 square inches of opening for each 100 amp. frame size, and 10 square inches for each 250, 400, or 800 amp. frame size circuit breaker, using the maximum requirement for any horizontal row of vertical circuit breakers in the compartment if horizontally mounted, this requirement shall apply for each circuit breaker. Ventilating louvers and grill openings shall be designed to minimize probability of entrance of solids and shall protect personnel against injury from electric arcs.

3.8.10 Vibration. - Units of all switchgear groups shall be designed and constructed to withstand the type I vibration test as specified in MIL-STD-167. The number of units to be tested shall be as specified in 4.3.2.

3.8.11 Materials. - The switchboard structure may be either of steel or aluminum construction as specified (see 6.1). Steel conforming to QQ-S-741 (grade C) or ASTM-A-131 shall be utilized in the fabrication of switchboard structures. Steel conforming to QQ-S-698, QQ-S-700 or ASTM-A-415 shall be used for plates and sheets. Aluminum alloy conforming to temper H111 of QQ-A-200/5 or temper T6 of QQ-A-200/8 shall be used for structural pieces. Aluminum alloy conforming to temper H32 or H34 of QQ-A-250/8 or temper T6 of QQ-A-250/11 shall be used for plates and sheets. Aluminum alloy conforming to temper H22 and H24 of QQ-A-250/2 may be used for nonstructural applications. Aluminum alloy conforming to temper 0 of QQ-A-200/5 shall be used for structural pieces which are bent into shape.

3.8.12 Thread locking devices. - Locking devices shall be provided on the following basis for electrical connections and mechanical assemblies in all switchboards and switchgear units, and in all equipment unless otherwise specified in the individual equipment specifications.

3.8.12.1 Electrical connections. - All nuts, bolts, studs, and screws used for electrical connections shall be secured by means of an acceptable locking device, except that a locking device need not be provided where solderless type connectors are used for conductors below 4000 circular mils (cm).

3.8.12.2 Mechanical assemblies. - All nuts, bolts, studs, and screws used for structural purposes shall be secured by means of an acceptable locking device where continuous satisfactory operation, under the shock vibration, and heat conditions specified herein, or as specified in the individual equipment specifications, depends on maintaining tight connection of parts, or where a holding screw, nut, bolt, or component part could fall into the equipment.

3.8.12.3 Acceptable locking devices. - The following are considered acceptable types of locking devices:

(a) Nut and lock nut.
(b) Self locking nut.
(c) Castellated nut with cotter pinning or safety wiring.

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(d) Split ring lock washer.
(e) External toothed lockwasher may be used where the weight of the part does not exceed 2 ounces per screw.

3.8.13 Welding. - Welding and allied processes shall be in accordance with MIL-STD-278.

3.9 Insulating bases, and bus supports. -

3.9.1 Insulating material. - In order to restrict the use of material which might catch fire or emit fumes when exposed to flame, the quantity of insulating material shall be kept to a minimum and used only where necessary to provide electrical insulation between live parts and to ground. Insulating material for bus supports and spacers, fuse bases and other similar parts shall be type GME of MIL-P-15037.

3.9.2 The design, number, spacing, and mounting of the bus supports shall be in accordance with calculations based on the magnitude of the short circuit stresses which may occur on the buses being supported.

3.10 Hinged and removable enclosing panels. -

3.10.1 Structure. - The panels shall be made from sheet 3/32 inch to 1/8 inch thick of U. S. Gage number 11. (Commercial tolerances acceptable.) The panel shall be formed with bent angle or channel edges. The flanges of all panels shall be welded at the corners and ground smooth. There shall be no butt joints on the panel surfaces. Where the strength of the panel is materially weakened by cutouts, it shall be reinforced by suitable stiffening members. The front enclosure panels for all type AQB and NQB feeder circuit breakers shall be drilled with 5/16-inch diameter holes so that the contact prong of the megger testing equipment may be inserted to make contact with the circuit breaker terminal for making the insulation tests of the feeder cables. Type NQB and AQB circuit breakers that are interlocked to form a bus transfer switch, casualty power circuit breakers and motor operated type AQB circuit breakers or any AQB circuit breakers which may have their load side energized from another source shall not be provided with megger holes. The size of the cutout for type AQB and ACB circuit breaker operating handles and escutcheons shall be in accordance with MIL-C-17361 and MIL-C-17587, respectively.

3.10.2 Panel hinging. - Panels shall be hinged where necessary to provide access to the back of the equipment mounted thereon and to equipment mounted within the switchboard frame assembly. Means shall be provided so that these hinged sections may be opened without removing the guard rail. Hinged panels shall be provided with door stops or positioning devices to stop the door in its opening swing, and to hold it in the open position. Panels shall be hinged on the left-hand side when facing front of the panel. A typical panel hinge assembly is shown on figure 8.

3.11 Mounting of equipment. -

3.11.1 Securing of equipment. - The equipment shall be adequately secured to the panels of framework so as to insure against dislodgment due to shock. Brackets and mounting lugs or feet shall have fully enclosed holes or slots. Open end slots will not be acceptable.

3.11.2 Mounting bolts and screws. - The bolts used in the assembly of the framework and for securing the large appurtenances to the framework shall be hexagon head type. Screws for miscellaneous purposes shall be round, fillister or flat slotted head type.

3.11.3 Arrangement of equipment. - No apparatus shall extend beyond the edges of the panel on which it is mounted. Symmetrical arrangements of equipment on the units is desirable, but should not take precedence over arrangements where space may be saved for future or spare equipment.

3.11.4 Polarity. - For direct current, when facing the front of the switchboard or panels, all left-hand contacts shall be of the same polarity (negative) and all right-hand contacts of the same polarity (positive).

3.11.5 Phase rotation. - For alternating current, when facing the front of the switchboard or panel, the phase rotation shall be A, B, and C, respectively, from right to left, top to bottom, or front to back. In special cases where space is a problem, phases may be transposed when specifically approved by the command or agency concerned.

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3.11.6 Casualty power equipment. -

3.11.6.1 Arrangements. - For ships service and emergency switchgear groups, the casualty power circuit breaker and riser terminal (required herein on the generator units) must be available for use in the event of damage to the remainder of the switchboard on which the generator unit is located. The lead from the generator bus on generator units for the main bus on power distribution units to the casualty power connection and equipment shall be as short and direct as practicable, and the casualty power terminal shall be readily accessible to permit rapid connection of the casualty power cable.

3.11.6.2 Mounting of casualty power equipment. - The casualty power circuit breaker and riser terminal shall be mounted at the rear or on the side of the unit. When mounted within the unit, the casualty power equipment shall be readily accessible from the rear of the unit so that cables may be connected to the terminal without undue exposure of personnel to live parts. The rear enclosures shall not be used for support of the terminal or circuit breaker, nor shall its removal be necessary for access to the terminal or for operation of the circuit breaker. When mounted on the side of the unit, the riser terminal shall be surface mounted and the circuit breaker shall be removable without removal of the side sheet. Normally, the circuit breaker and riser terminal shall be located adjacent to one another so that personnel may readily determine that the breaker is open and the terminal not energized before making connections. Where adjacent locations are not practicable, the terminal shall be enclosed to avoid undue exposure of personnel to live parts. Tools for riser terminals shall be mounted in a rack on the outside of the switchboard enclosure adjacent to the terminal.

3.11.6.3 Casualty power circuit breaker rating. - The circuit breaker type AQB-A or AQB-LF, shall be provided with an element rated at 250-amperes. Instantaneous trip settings shall be as follows: AQB-A circuit breaker — 3400 amperes, AQB-LF circuit breaker 3000 amperes. For generators rated 1000 kw. and below, an AQB-A circuit breaker shall be furnished unless otherwise specified in the contract or order. For generators rated above 1000 kw., an AQB-LF fused circuit breaker shall be furnished.

3.11.7 Voltage regulator equipment. - The space and arrangement of voltage regulator equipment shall be based on permitting replacement, with the minimum modification of mounting and arrangements of voltage regulators of one manufacturer's type with those of another manufacturer's type (see MIL-R-2729).

3.12 Insulation distances. - Insulation distances in air, and surface creepage distances on insulating materials for all appurtenances shall conform with the applicable specifications. For all other live parts, such as bus work, the values shall conform to table I, except the applicable clearances of MIL-E-917 may be used where the bus is connected to the component studs.

Table I - Switchboards.

Voltage	Insulation distances to ground or to opposite polarity		
	In air	Surface creepage	
		Bottom and side surfaces	Top-flat surfaces 1/
	Inches	Inches	Inches
125 a.c. or d.c.	1/2	15/16	1-1/8
230 a.c.	5/8	1-3/16	1-1/2
500 a.c. or d.c.	25/32	1-9/16	2
1000 a.c.	1-5/16	2-1/2	3-11/32

1/ Values for top creepage apply to flat surfaces or to curved surfaces presenting sufficient irregularities to permit accumulation of dust and moisture. Values are not intended to apply to simple cylindrical surfaces having a radius of 3 inches or less where side creepage values will apply.

3.13 Bus bars and connectors. -

3.13.1 Bus connections. - All bus connections shall be made with solid copper or aluminum alloy bus bars, except as specified for certain circuit breaker connections.

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3.13.2 Copper bus bars. - Rectangular copper bus bars with rounded corners or round edges, hard temper in accordance with ASTM B-187 shall be used for bus bars. The bars shall be silver surfaced at the contact areas as specified herein.

3.13.3 Cable connector sizes. - On switchboards having an a.c. bus structure designed for short circuit currents of 15,000 amp. or less, single conductor insulated copper cable, type D4 of MIL-W-16878/3, shall be used for connections to 100 amp. frame size type AQB circuit breakers. On switchboard buses having a total connected d.c. generator capacity of not more than 150 kw. at 120 V., similar cable connections shall be provided for 100 amp. frame circuit breakers.

3.13.4 Aluminum bus bars. - Rectangular aluminum alloy 6101, T61 or T62 temper with rounded corners or round edges in accordance with ASTM B-317 shall be utilized for bus bars. The bars shall be silver-surfaced at the contact areas as specified herein.

3.13.5 Cable connector sizes for casualty power. -

3.13.5.1 Cable connecting a casualty power circuit breaker to the line side of an a.c. generator circuit breaker shall be type D 1/0 of MIL-W-16878/3. Cable connecting the casualty power circuit breaker and the riser terminal shall be type D-4.

3.13.6 Bus bar sizes. - The size of the bus bars shall be selected on a basis of the current carrying ratings shown in table II. In no case, shall the bus bar sizes be smaller for a given current than the corresponding sizes listed in table II. The ratings shown are based on a 50° C. ambient and on the separation or arrangement of phases sufficient to neglect mutual inductance between phases and interference with radiation of neighboring conductors. If the spacing is not large enough to neglect mutual inductance between phases or interference in radiation, a reduction in the values given in table II shall be made.

Table II - Ampere rating of rectangular bus bars placed on edge.

Number of bars in parallel	Size of bars (inches)	Cross sectional area, (square inches)	A.c. ampere rating 60 cycle		D.c. ampere rating
			Copper bus, silver surface	Aluminum bus silver surfaced	Copper bus, silver surface
1	1/2 x 1/8	0.063	140	100	140
	5/8 x 1/8	.078	175	120	175
	3/4 x 1/8	.094	210	140	210
	1 x 1/8	.125	285	190	285
	1-1/2 x 1/8	.188	425	280	425
	2 x 1/8	.250	555	370	555
	3/4 x 3/16	.140	265	175	265
	1 x 3/16	.188	355	235	355
	1-1/2 x 3/16	.278	550	360	550
	2 x 3/16	.375	700	450	710
	3/4 x 1/4	.188	295	200	295
	1 x 1/4	.250	410	300	410
	1-1/2 x 1/4	.375	600	420	600
	2 x 1/4	.500	780	550	800
	2-1/2 x 1/4	.625	1,000	660	1,050
	3 x 1/4	.750	1,140	800	1,185
	4 x 1/4	1.000	1,425	1,000	1,490
	5 x 1/4	1.250	1,760	1,250	1,850
	6 x 1/4	1.500	2,100	1,500	2,190
2 (1/4 inch apart)	1 x 1/4	.500	650	480	650
	1-1/2 x 1/4	.750	950	660	950

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Table II - Ampere rating of rectangular bus bars placed on edge (Cont'd.)

Number of bars in parallel	Size of bars (inches)	Cross sectional area, (square inches)	A.c. ampere rating 60 cycle		D.c. ampere rating
			Copper bus, silver surface	Aluminum bus silver surfaced	Copper bus, silver surface
2 (1/4 inch apart)	2 x 1/4	1.000	1,350	900	1,370
	3 x 1/4	1.500	1,825	1,250	2,000
	4 x 1/4	2.000	2,280	1,600	2,530
	5 x 1/4	2.500	2,740	1,900	3,100
	6 x 1/4	3.000	3,140	2,250	3,630
3 (1/4 inch apart)	3 x 1/4	2.250	2,200	1,600	2,620
	4 x 1/4	3.000	2,660	2,000	3,110
	5 x 1/4	3.750	3,200	2,400	3,830
	6 x 1/4	4.500	3,600	2,800	4,560
4 (1/4 inch apart)	3 x 1/4	3.000	2,650	-----	3,130
	4 x 1/4	4.000	3,020	-----	3,870
	5 x 1/4	5.000	3,450	-----	4,750
4 (2 pairs/phase 3/4 inch between pairs)	6 x 1/4	6.000	4,000	-----	-----
4 (2 pairs/phase 2-1/2 inches between pairs	5 x 1/4	5.000	4,200	3,200	-----
	6 x 1/4	6.000	5,000	4,000	-----

3.13.7 Grouping and arrangement of bus bars. - Particular attention shall be given to the grouping and arrangement of buses for a.c. (60-cycle and above) currents near magnetic material, since proximity of such material will create additional heating, thereby interfering with the dissipation of heat from conductors and reducing their capacity.

3.13.8 Bus bar stresses. - The bus work assembly shall be designed to withstand the stresses resulting from the maximum short-circuit currents which might be encountered under normal operation of the electric plant. Conditions of high impact shock which may be encountered aboard the ship shall not mechanically damage the bus work nor reduce the bus clearances below minimum requirements. Where single conductor cable is used for connection to circuit breaker, the cables shall be securely bound together with noninflammable cord and secured to the supports as necessary to prevent distortion under short circuit conditions. The binding shall be applied in such a manner that it will not cut into the cable insulation due to the forces exerted under short circuit conditions. The binding shall be especially heavy where cables turn out of the pack. Terminal lugs shall be used for terminating cable at the circuit breaker terminals and at the bus bar. Through-bolts shall be used for securing the cable terminal to the bus bar. Not more than two cable terminals shall be clamped by one bolt, and when two are used, they shall be placed on the opposite sides of the bus. The cables shall be neatly formed, and where their length exceeds approximately 12 inches, they shall be supported by the switchboard structure.

3.13.9 Bus bar forming. - Flat bends shall have an inside radius of not less than the thickness of the bus bar, and the ends of the bus bars shall be neatly finished. Edgewise bends of bars up to 2 inches in width, may be made on an inside radius of 1 inch, and those over 2 inches, but not exceeding 4 inches in width, may be made on an inside radius of 2 inches. Bars of greater width should not be bent edgewise. Bars shall be free from cracks or flaws at bends.

3.13.10 Silver surfacing of buses. -

3.13.10.1 The silver surfacing of buses and connections shall be accomplished by plating or by amalgamation process using 99.9+ percent pure silver. The silver thus applied shall not be less than 0.0002 inch thick and shall satisfactorily withstand adhesion tests specified in QQ-S-365, and shall show no tendency to

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peel. The contact surfaces of all bus bars shall be silvered up to 1 inch past the joint area. In lieu of this, on copper bus bars, only an area around each hole may be silvered and the silvered area around the hole shall be not less than 1/8 inch wide band. The entire bus may be silver surfaced at the discretion of the manufacturer. Threaded surfaces shall have silver thickness of at least 0.0002 inch.

3.13.10.2 Holes in bus bars for bolting may be either punched or drilled, but the contact area shall be smoothed by belt sanding, or other equivalent method, prior to silver surfacing. Where punching is used, the operation shall not appreciably indent the surface of the bus bar in the vicinity of the hole. Slotted holes are not permitted. The contact surfaces of bus bars and contact nuts shall be finished true to give adequate contact. Contact nuts and contact spacer nuts for copper stud contacts shall be of rolled brass and shall be of sufficient size to provide the necessary area for contact surface. All contact nuts and contact spacer nuts for silvered contacts shall be silver-surfaced, as specified in 3.13.10.1, for threaded surfaces. Where carriage bolts are used, square holes with the following dimensions shall be used:

Bolt size (inch)	Hole size (inch)	Radius of hole corner (inch)
3/8	7/16	1/32
1/2	19/32	1/16
5/8	23/32	1/16

Holes in bus bars for standard bolts shall have the following dimensions:

Bolt size (inch)	Hole size (inch)
3/8	7/16
1/2	9/16
5/8	11/16

3.13.10.3 The joints of all bus bars shall be specially prepared before bolting. The method of preparation shall be as follows:

(a) Apply a thick film of high viscosity paste conforming to MIL-C-16173.
(b) Bolt joints together without removing the paste.
(c) After joints are bolted together, remove paste from all exposed surfaces with clean dry cloth.
(d) On aluminum bus bars where only the joint or contact area is silver plated the paste shall be applied approximately 1 inch beyond the silver plated area.

3.13.11 Bolting of bus bars. - Bolts and nuts used in bus bar joints shall be of steel, grade 2 of MIL-B-857, coarse thread series, and suitably plated to protect against corrosion. Flat washers shall be of similarly plated steel and shall be used under all bolt heads and nuts adjacent to the conductor. Securing bolts shall be fitted with nuts conforming to figure 9. For typical bus bar mountings see figure 10. Locking devices listed in 3.8.12.3(b), (d), and (e) are not to be used in bus bar joints. Carriage bolts may be used for bus bar joints. Where carriage bolts are used the flat washer under the head shall be omitted. Self-locking nuts with nylon inserts may be used.

3.13.12 Contact pressure of bus bars. - Bus bar joints shall be made in accordance with table III and figure 9. Bolts for bus bar joints shall be lubricated and the following minimum torques shall be used:

Bolt size Inches	Torque Inch - pounds
3/8	170
1/2	360
5/8	600

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Table III - Bolting of connection bars - size and number of bolts.

Size bars	Size of bars (inches)						
	3/4	1	1-1/2	2	3	4	5
Inches	Inch	Inch	Inch	Inch	Inch	Inch	Inch
3/4	One 3/8	-------	-------	-------	---------	--------	--------
1	One 3/8	One 1/2	-------	-------	---------	--------	--------
1-1/2	One 3/8	One 1/2	One 5/8	-------	---------	--------	--------
2	Two 3/8	Two 3/8	One 5/8	One 5/8	---------	--------	--------
3	Two 3/8	Two 1/2	Two 1/2	Two 1/2	Two 5/8	--------	--------
4	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Two 5/8	Four 5/8	--------
5	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Three 5/8	Five 5/8	Five 5/8
6	Two 3/8	Two 1/2	Two 5/8	Two 5/8	Three 5/8	Five 5/8	Five 5/8

3.13.13 Spacing of bus supports. - The bus work shall be designed to withstand the maximum root mean square (r.m.s.) asymmetrical short circuit current. For the calculation of required spacing of bus supports to withstand the short circuit current, the value of maximum stress in the outside fibers shall be based on the yield strength of the bus bar and not upon the tensile strength.

3.13.14 Bus bar capacity. - Main bus rating shall be equal to that required for standardized units as shown on Drawing 9000-S6201-74453. Bus taps off of the main bus shall either have a rating equal to the main bus or have a rating equal to the sum of the circuit breaker frame sizes that they supply, whichever is less. For individual circuits, the bus connections shall have a capacity based on the frame size of the circuit breaker.

3.13.15 Cable connector types. - Cable connectors installed on all cables entering or within the switchboard shall be of the solderless type, in accordance with MIL-E-16366. Type CLSS cable connectors shall be used with type AQB circuit breakers except where multiple cables are used type CLS cable connectors may be used. Screw or crimp type cable connectors shall be used with type ACB circuit breakers or for bus bar applications. When using crimp type lugs the manufacturer of the lug shall be the same as the manufacturer of the crimping tool to be used by the installing activity. Cable connectors shall not be installed on spare circuit breakers to which a cable is not attached. For all cables of 4000 cm. and larger, cable connectors shall be secured by means of locking devices in accordance with 3.8.12.3. Cable connectors for cables entering a switchboard may be furnished by the installing activity (see 6.1).

3.13.16 Connections between buses. - Connections between the buses of adjacent sections of a switchgear group are made by cables; instead of solid bus bars, to permit relative movement of the sections under shock without rupturing the switchboard main bus circuit. A means for readily disconnecting the cables from the bus shall be provided in all sections having a source of power other than from the connecting cables. The disconnecting device shall have a rating equal to the main bus. The cable connections shall be made by the installing activity. The switchboard design arrangement shall include means to provide for the entrance of the connecting cables into the switchboard structure, and for connecting the cables to the switchboard bus. The connecting cables shall have a current rating equal to that of the smaller of the two connected buses.

3.13.16.1 Consideration shall be given to facilitate connection (by the installing activity) of the ship's cables to the switchboard so that these cables will not result in restriction of access features. Armor will be removed from the portion of ship's cables inside the switchboard. Instructions for interconnecting wiring shall be shown on the switchboard drawings.

3.13.17 Cables entering switchboards. - Cables will enter the switchboard through the top or bottom drip shields with standard terminal tubes (see 6.1). The terminal tubes are to be shipbuilder furnished. The switchboard design arrangement shall provide for proper cable connections without restriction of access features.

3.14 Wiring of control and instrument circuits. -

3.14.1 Control and instrument cable types. - The cables used for control and instrument circuits, ground detector and indicator lights and other parts, shall conform to type D of MIL-W-16878/3. Type D-16(19) wire

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shall be used where the rated current carrying capacity is adequate for the circuit application. NAVSHIPS 362-2352 shall be used as a guide for the rating of bundled cables. The flexible 19 strand wire shall be used for all wiring between hinged panels and the stationary structure. All cables within each unit shall be of the same solid color.

3.14.2 Cable end connectors. - The ends of each wire shall be connected to the apparatus studs by means of approved type solderless terminals or by forming a wire loop around the apparatus stud or terminal screw and retaining the loop in a cup or crimped washer. The wire loop shall be soldered to secure the strands together. Locking devices in accordance with 3.8.12 shall be used to secure the nuts or screws connecting wire loops to the apparatus connection point. See figure 11 for illustration of wire connections.

3.14.3 Forming and securing cable groups. - The wires shall be neatly formed into groups which shall be laced or corded and supported or clamped in a manner which will prevent chafing of the insulation due to vibration. Commercial nylon cable straps or metal clamps may be used, but where metal is used it shall be covered with a flame retardant material which will prevent chafing of the wire insulation. Lacing material shall be either nylon straps, non-inflammable cord, glass fiber tape conforming to type SR-1, of MIL-I-3158, or pressure sensitive adhesive plastic tape, conforming to MIL-I-7798. The last two materials are specifically adapted for serving switchgear control wiring in locations subject to abrasion. Glass fiber tape may be coated with adhesive on one side, but when used with this adhesive coating, it shall be covered with pressure sensitive adhesive plastic tape for permanent adhesion. There shall be no splices in wire and all connections shall be made at the terminal studs of the devices or terminal boards. The wire groups running from hinged panels shall be formed and clamped so that sharp bends do not occur with the panel in either the open or closed position and shall be connected to terminal blocks on the fixed portion of the switchgear structure. Where wires run from one section of a switchboard to an adjacent one, and these sections are separated for shipping or installation purposes, terminal boards shall be provided on one of the units and located so as to facilitate completion of the interconnections during installation of the switchboard on the vessel.

3.14.4 External control cable connections. - External control cable connections to the switchboards shall terminate at terminal boards accessibly located near the top or bottom of the switchboard, corresponding to ship's cable entrance. Where calibrated shunt leads are required, they shall terminate on the instrument studs. An excess length of shunt leads shall be neatly coiled and secured so as not to interfere with access to other equipment. Small wiring to circuit breaker trip coils shall run direct to the coils and associated contact circuits and shall not be routed through terminals of other apparatus. Taps from bus bars shall be made by bolting terminals to the bus bars.

3.14.5 Synchronizing control circuits. - The synchroscope shall be supplied by the phase A-B transformers, and the synchronizing lights by the phase B-C transformers provided for metering, synchronizing, and relaying potential as shown on figure 12. The loss of one transformer shall not disrupt both synchronizing facilities. The closing control circuit of electrically operated circuit breakers (a.c. generators, bus ties and shore power) shall be connected through contacts of the corresponding synchronizing switch. Synchronizing protective equipment shall be provided as specified in MIL-S-24188 to electrically prevent closing of the generator circuit breakers, bus tie circuit breakers, or shore power circuit breaker, as applicable, unless the energized systems to be connected are properly synchronized.

3.14.6 Space heater circuits. - When space heaters are provided with the associated generator, the heater circuit shall be supplied through a heater control switch from a local lighting circuit. An indicator light (white) shall indicate when generator heaters are energized. This heater circuit shall be wired through a "b" switch of the associated generator circuit breaker.

3.14.7 On surface ships, the indicator lights of the mimic bus shall be supplied from a local lighting system which has available both normal and emergency power. On submarines, power for the indicator lights shall be taken from the a.c . or d.c. control power buses.

3.14.8 The control transformers shall be supplied in number as required for supplying governor motor control circuits, temperature indicator, and generator running and up to voltage relay.

3.14.9 Ground detector lights shall be connected as shown on figure 13.

3.14.10 The voltage regulator equipment shall be connected in accordance with the master drawings provided with the regulator equipment.

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3.14.11 Wire and terminal marking. - All internal wiring of equipment shall be marked on both ends by fiber tags, or by synthetic resin tubing, type F, grade A of MIL-I-631. Markings shall agree with the designation shown on the approved switchboard drawings to facilitate checking of connections. It will not be necessary to mark the terminal studs on instruments, and control switches, but wiring diagrams shall clearly indicate the studs or contacts to which marked wires are connected. Terminal blocks shall be equipped with marking strips. These marking strips shall be permanently marked at each terminal with a number (i.e., 1 to 12) and also to agree with the corresponding number on the wiring diagram. Terminal blocks in each unit shall be numbered TB1, TB2, and so forth.

3.14.12 Shunt trip coil circuits (both a.c. and d.c.) for tripping type ACB and AQB circuit breakers shall be routed through a normally open contact (open when the breaker is open) of the circuit breaker auxiliary switch, so that the trip coil cannot remain energized after the circuit breaker is open.

3.15 Circuit breakers. - Circuit breakers shall conform to MIL-C-17587 (for type ACB), MIL-C-17361 (for types AQB and NQB), and MIL-C-17588 (for type ALB). Type ACB circuit breakers 1600H amp. frame size and larger shall be electrically operated and circuit breakers 1600 amp. frame size and smaller shall be manually operated unless otherwise specified. All circuit breakers controlled from a benchboard, or controlled by the synchronizing protective equipment specified in 3.14.5., shall be electrically operated.

3.15.1 Mounting (AQB and NQB). - The types AQB and NQB circuit breakers for switchboard use shall be equipped with removable type contacts on both line and load side, connections being made by use of terminal mounting blocks in accordance with MIL-C-17361. Distribution type, horizontal panel mounting of AQB circuit breakers is permitted in accordance with Drawing 9000-S6201-74453. When circuit breakers are mounted vertically, the line side terminals shall be at the top. When interlocks are required for types AQB and NQB circuit breakers, they shall be front of panel mounted and shall be suitable for use with circuit breakers of any make by no further modification than possible reassembly of the interlock.

3.15.2 Mounting (ACB). - Type ACB circuit breakers shall be of a removable assembly design. Each circuit breaker shall be supplied as a complete removable switchboard assembly consisting of circuit breaker, separable disconnects and associated control wiring, circuit breaker drawout mechanism and assembly structural supports and back plate including stationary main bus connections and stationary terminals for connection of the necessary external control wiring.

3.16 Switches, disconnecting devices, and transfer switches. -

3.16.1 All switches (control, instrument) shall be in accordance with MIL-S-18396 or MIL-S-15291.

3.16.2 All instrument and control switches shall have gray or natural light tan escutcheon plates with black lettering. The switches shall be provided with four styles of handles to identify the function for which the switch is provided. The handle styles and corresponding function shall be as follows:

(a) Round - governor control. 1/
(b) Oval - synchronizing and voltage regulator.
(c) Round (knurled) - instruments (voltmeter, ammeter, frequency meter, temperature indicator); test 1/ (emergency switchboard).
(d) Pistol-grip circuit breaker 1/, ground lamps 1/, control bus selector.

1/ Shall have spring return handles.

3.16.3 All bus disconnecting switches for performing power disconnecting functions shall be of the type in which contact surfaces are forced together by pressure applied by means of screw threads. With full rated current through the disconnect devices, the maximum temperature rise of the devices shall not exceed 50° C. The disconnect devices shall be mounted inside the switchboard and shall be readily accessible after removal of the rear enclosure. Tools (if required) for operation of the devices shall be mounted outside of the switchboard enclosure adjacent to the panel that must be removed for operation of the device. A warning plate shall be provided near the disconnect device to read: DO NOT OPERATE WHEN ENERGIZED. The plate may be constructed in accordance with the requirements of 3.22.4 with letters not less than 3/16 inch.

3.17 Fuses and terminal blocks. -

3.17.1 Fuses shall be in accordance with MIL-F-15160. Identification plates for current limiting fuses shall state that these fuses shall be replaced only with current limiting fuses or a warning plate clearly pointing this out shall be added.

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3.17.2 Fusing of transformers. - Each metering potential transformer shall be fused on the primary side with a separate 6 amp. fuse in each leg of each transformer. Control transformers shall be fused on the primary side. Transformer secondary circuits shall not be fused, except where a circuit leaves the switchboard. In such a case, the ungrounded portion of the circuit which leaves the switchboard shall be fused, and the fuses shall be located so as not to open any circuits on the local switchboard. One 6 amp. fuse shall be provided in each leg of each indicator light transformer except in those cases where protection is provided by other fused circuits. The control circuits for electric operation of circuit breakers shall be fused for short circuit protection only. Potential circuits for generator voltage regulators on excitation systems shall not be fused.

3.17.3 Fuse mounting. - The fuses mounted on switchboards shall be accessibly located for replacement (see figure 14). Dead front fuseholders or clip type fuseholders may be used. The fuses shall be grouped on subpanels insofar as practicable, but the length of leads between fuses and the power circuit taps shall be kept to a minimum.

3.17.3.1 Dead front fuseholders shall be of the following types and shall not be installed for any distribution circuit fuse on power or lighting distribution switchboards or panels:

Type	Specification
FHL10U	MIL-F-19207/1
FHL11U	MIL-F-19207/2
FHL12U	MIL-F-19207/3
FHL32W	MIL-F-19207/21
FHL33W	MIL-F-19207/22
FHL34W	MIL-F-19207/23
FHL35W	MIL-F-19207/24

3.17.3.2 Clip type fuse holders shall be enclosed by hinged metal covers. The covers shall be provided with insulating strips which will hold the fuses in the fuse clips under shock. Fuse clips shall be in accordance with MIL-C-21346/1.

3.17.4 Terminal boards. - To minimize the possibility of breakage of the terminal boards, due to distortion of the mounting support under shock the length of the terminal board shall not exceed 12 terminal points.

3.18 Instruments, relays and instrument transformers. -

3.18.1 Electrical measuring instruments. - Electrical measuring instruments shall conform to MIL-I-1361, MIL-T-2867, MIL-M-16034, MIL-I-16104, MIL-M-16125, MIL-W-19088, MIL-V-23151 and MIL-M-23167, having a 4-1/2 inch, rectangular case with 250-degree scale. Instrument scales shall be white with black markings and lettering, and shall be marked to indicate the full load and normal voltage calibration point. Figure 15 shows typical mounting of instruments.

3.18.2 Instrument transformers. -

3.18.2.1 Instrument transformers for metering and relaying shall conform to MIL-I-1361, having an accuracy for Navy secondary standard (N.S.S.) rating. Transformer type devices used to provide phase angle changes for single phase wattmeters on three-phase circuits or similar functions shall conform to MIL-I-1361.

3.18.2.2 Instrument transformers provided for voltage regulators shall not be used to supply any other burden. For each group of instruments, relays, and synchroscope, one set of instrument transformers shall be used unless the burden on one set would be too great for the accuracy required. The potential transformers for each group of instruments, relays and synchroscope shall not be used for other potential devices without approval by the command or agency concerned. Where two potential transformers are used for three phase operation, they shall be connected open delta on the primary and secondary. One transformer shall be connected across phase A-B and the other across phase B-C.

3.18.2.3 Where current transformers are required to supply 3-phase current, two transformers shall be used, one connected in phase A, one in phase C. Their secondaries shall be connected open delta. Current transformer secondary circuits which extend beyond the section containing the current transformers shall be provided with protective devices conforming to MIL-I-1361, to prevent high voltage in the event of an open circuit. The protective devices shall be connected so that their operation will not shunt out protective relays

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which may be connected in the circuit. A short circuiting switch shall be connected in parallel with the protective device for manually short circuiting the remote part of the current transformer circuit. Connections shall be as shown on figure 16.

3.18.2.4 One secondary terminal of each metering and relaying (potential and current) transformer shall be grounded, except for 3-phase connection, where one ground connection shall be made to the common or B phase of the two transformers.

3.18.3 All relays, both protective and auxiliary types, shall conform to MIL-R-2033 and shall be of types which have been tested and approved for class HI shock resistance. For auxiliary relays, also see MIL-R-19523 and MIL-C-2212.

3.18.4 Control transformers shall conform to MIL-T-15108 and MIL-T-16315, as applicable.

3.18.5 Resistors and rheostats shall conform to MIL-R-15109.

3.19 Indicator lights. -

3.19.1 Indicator lights shall conform to Drawings 9000-S6202-73907 for a.c. and 9000-S6202-74307 for d.c. All indicator lights shall be furnished complete with lamps.

3.19.2 For position indication of circuit breakers, blue indicator lights shall be used to show that the circuit breaker is closed. All other indicator lamps shall have color caps as specified for the particular application. The color code shall be as specified in table IV.

Table IV - Globe color designations.

Red	Danger or emergency
Green	Normal
Clear, etched (white)	Power on or power available
Blue	Closed
Clear (not etched)	Synchronizing and ground detector lights
Amber	Abnormal
Yellow	Open

3.20 Switchboard enclosures. -

3.20.1 Rear enclosures. - The rear of each unit shall be provided with a removable enclosure. These rear enclosures shall be made of expanded metal (rolled flat) perforated sheet or solid sheets. Rear enclosures shall be sectionalized into not less than two horizontal parts to facilitate handling. When solid sheets are used ventilation openings shall be provided at the top and bottom with louvers or openings similar to those in expanded metal or perforated sheet. The enclosures shall be adequately secured but shall not be used for structural strength. The openings in expanded metal or perforated sheet shall not be used for structural strength. The openings in expanded metal or perforated sheet shall not pass a rod having a diameter greater than 3/8 inch. Provision shall be made for removal of fuses without removing the rear enclosure of a unit. On bench type control units having equipment mounted on the rear enclosures the rear enclosures shall be solid sheet. Clearances between live bus work and disconnect devices, or both, and rear enclosures shall be adequate, in order to prevent these non-structural strength panels from coming into contact with live bus work, if the panels are buckled inward.

3.20.2 Top enclosure. - The top of each section shall be enclosed by a sheet cover. The top shall be of suitable design to prevent dripping water or falling objects from entering the switchboard enclosure or damaging instruments on the front panels. It shall extend approximately 4 inches out over the face of the front panels and 4 inches beyond the rear of the switchboard structure. The clearance between the upper edge of the front panels and the under side of the top sheet, where it extends over the front shall be at least one-half inch. The front and rear edges of the top sheet shall be turned up to form a 1-inch flange along the entire length of the section or switchboard to prevent water from spilling over the front or rear. To insure watertight integrity at the corners, the turned-up front and rear flanges shall be extended around the corners and along the end of the top sheet approximately 8 inches. The end edges of the top sheet shall be turned down to form a flange, and shall overlap the side sheets of the switchboard to allow drainage over the end in a manner which prevents water from seeping down the inside of the side sheet.

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3.20.3 Panel fastening. - Hinged and removable panels on the front and rear of the unit shall be secured by thumb screws made captive in the panel (see figures 17 and 18). The method employed to make thumb screws captive shall be of sufficient mechanical strength to prevent failure which could allow metal parts to fall into the switchgear when subjected to conditions of repeated usage. Single-thread engaging nuts formed by stamping a thread-engaging impression in a flat piece of metal shall not be used.

3.20.4 Protection of enclosure from waterspray for submarine applications (see 6.1). - All switchgear covered by this specification which supply services vital to the safety of the submarine shall be protected from waterspray from any angle and shall continue to operate when subjected to the waterspray test requirements of 4.3.5. Protection may be obtained by means of transparent, gasketed, quick opening covers mounted over circuit breaker handles, fuses, or other appurtenances subject to water damage and by installing baffles over louvers to exclude waterspray. An operator's view of instruments and his ability to operate equipment shall not be impaired. Forced draft ventilation shall not be used for any electrical equipment.

3.21 Guard rails. -

3.21.1 Each switchboard unit, except for distribution units, shall be provided with a vertical grab rod located on the right hand side, facing the unit (see figure 19). The rod shall be of smooth hardwood, supported with a sufficient number of brackets to provide adequate support for personnel under all operating conditions of the ship. The rod shall be attached in such a manner that all panels can be removed and hinged panels opened, without removal of the rod from the member to which it is attached. These rods shall be located so that they will not interfere with operation of the equipment and shall not extend more than 9 inches from the front of the panel.

3.21.2 Removable guard rail sections shall be provided on the rear framework of the switchboard (see figure 20). In general, two horizontal rails will be required at proper levels to prevent personnel from being thrown against live buses, due to the motion of the ship. They shall be mounted on the inside of the rear vertical members so as not to protrude beyond the rear or ends of the structure. These guard rail sections shall be easily removable for maintenance access but shall be provided with securing bolts or other locking means to prevent dislodgment due to shock.

3.22 Painting. -

3.22.1 General. - The exterior and interior surfaces of all enclosures shall be painted. The order of operations shall be as follows:

(a) Complete all fabricating operations, such as welding, machining, drilling, and tapping.
(b) Remove all rust and other visible corrosion products.
(c) Remove all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.
(d) Apply primer pretreatment coating or chemical treatment and primer.
(e) Apply enamel.

3.22.2 Primer. - The primer shall be applied in accordance with either of the following two methods:

(a) One coat of primer pretreatment coating in accordance with MIL-C-15328, applied as a continuous film 0.0002 to 0.0005 inch thick.
(b) For steel, a phosphate treatment conforming to types I and II of TT-C-490 and for aluminum a phosphate treatment conforming to type I of TT-C-490 or a pretreatment conforming to MIL-C-5541, followed by one coat of primer in accordance with TT-P-664, TT-P-645 or MIL-P-8585, applied as a continuous film 0.0004 to 0.001 inch thick.

3.22.3 Enamel. - One coat of gray enamel conforming to class 2, type II or III, (with TT-P-645 primer use type I) of MIL-E-15090 shall be applied as a continuous film, each approximately 0.001 inch thick. A second coat of enamel shall be applied to the outside of the switchgear.

3.22.4 Bus bars. - Copper and aluminum bus bars on switchboard shall have one coat of red colored insulating paint, conforming to MIL-I-17384, or other specifically approved paint. Coating shall be flexible, tough, moisture and acid resisting, and shall have an insulation resistance value of at least 500 volts per mil of thickness. One coat, 0.004 to 0.006 inch thick, shall be applied. Before painting, burrs and sharp edges shall be removed and bus bars cleaned to remove dust, grease, oil, or other foreign material. Paint may be

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applied by spraying or may be flowed on with soft bristle brush when spraying is impracticable. Coatings shall be free from pinholes. Bus bars shall be painted before assembly up to one-quarter to one-half inch from the joint area. Where a painted area of a bus bar would be an inch or less in width, it need not be painted. Do not paint bus joints or connectors either before or after assembly. Taping of bus bars located entirely within the switchboard enclosure is prohibited. In lieu of the above, the bus bars facing and closest to the rear of the unit may be stenciled "DANGER 450 VOLTS". The applicable voltage may be substituted for 450 volts. The letters shall be red and not less than 3/4 inch in height. If the bus bar arrangement is such that letters are not clearly visible, the bus bars shall be painted. As an alternate to stenciling, the bus bars may be provided with a warning plate "DANGER 450V". The letters shall be not less than 3/4 inch high and shall be white on a red background. The plate shall be pressure sensitive adhesive backed elastomeric film with a carrier or separator sheet applied over the adhesive. The film shall consist of a white printed opaque polyvinyl fluoride film . 002 inch thick laminated to a . 001-inch clear polyethylene terephthalate film. The plate adhesive shall consist of an acrylic base vinyl polymer of a pressure sensitive type. A minimum of two warning plates per unit shall be provided; that is, one in the upper and one in the lower compartment.

3.23 Identification plates and marking. -

3.23.1 Identification plates shall conform to types A, B, C, F or H of MIL-P-15024. Plastic identification plates shall be light gray except beveled edges are not required. Inscription or markings for all type plates shall be black.

3.23.2 Location. - The identification plate shall be placed in close proximity to the operating handle of the equipment to which it refers, and generally either directly above or directly below it.

3.23.3 In general, identification plates shall be of a size as listed in MIL-P-15024. Other sizes of plates may be used when specifically approved by the command or agency concerned.

3.23.3.1 Thickness of identification plates shall be in accordance with MIL-P-15024.

3.23.3.2 Identification plates for rheostat handwheels and other rotary devices shall, where required, be circular or other shape to suit the equipment.

3.23.4 Identification plates with information consistent with the following shall be provided at or near the top of the front enclosure of each switchgear section or for each unit if procured individually:

(a) Name or functional designation of item.
(b) Manufacturer's name.
(c) Manufacturer's drawing number (front view).
(d) Year manufactured.
(e) NAVSHIPS drawing number (front view), if assigned.
(f) Stock number, if assigned.
(g) NAVSHIPS number of manual, if assigned.

3.23.5 Unless lack of space prevents, abbreviations of words on identification plates shall be avoided except in case of words which have abbreviations in common use in accordance with MIL-STD-12.

3.23.6 Installation. - Identification plates shall be fastened securely to such parts of the equipment as ordinarily will not be renewed during its service life. Identification plates shall be attached in accordance with MIL-P-15024. Metal identification plates and mimic bus may be attached with an adhesive conforming to MIL-A-22895.

3.23.7 Detail application requirements. - Identification plates shall be installed immediately above each circuit breaker and shall contain the following information:

(a) Circuit name.
(b) Circuit designation and number.

3.23.8 Identification plates without markings shall be provided for mounted spare circuit breakers.

3.23.9 Instruments shall be marked with the name of circuits; marking may appear on instrument or on a separately mounted identification plate.

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3.23.10 Fuses for each switchboard control circuit, instrument or instrument transformers shall have an identification plate showing name or designation of circuit, phase (if applicable) and fuse ampere rating and type designation.

3.23.11 Marking switchboard buses. - Buses shall be marked in a visible location with their polarity or phase identification.

3.23.11.1 Identification plates for buses may be similar to the warning plates specified in 3.22.4.

3.23.12 Positive, and negative polarities of direct current buses shall be indicated "+ and -" signs, respectively.

3.23.13 Phase identification of alternating current buses shall be indicated by the capital letters A, B, or C, as applicable. A neutral bus of a four-wire, three phase circuit, where used shall be indicated by the capital letter N. The letters A, B, C, in that order, shall indicate the phase sequence.

3.24 Drawings. - Drawings shall be submitted in accordance with MIL-D-1000/2. Type II drawings of individual units and type I drawings for complete switchboards shall be furnished. Delivery of originals, when required, and the type of reproductions shall be as specified (see 6.1).

3.24.1 Unit drawings. - Type II drawings of switchgear units shall cover the information specified in categories A, B, C, D, F and G of MIL-D-1000, including the following:

(a) Dimensional front, plan, side section and rear views of the switchboard unit, showing standardized overall dimensions of the unit, number, size and location of holes for interframe bolting, location of required equipment, location and size of main buses passing through the unit, and connections thereto. These views, together with the list of material, shall show the location and function of all appurtenances comprising the unit. The front and plan view of the unit shall, in addition, show all limiting dimensions including clearance necessary for hinged panels and all apparatus mounted on the front.
(b) A dimensional rear elevation view of the unit showing casualty power equipment and rear enclosures where specified.
(c) A dimensional floor or mounting view to show arrangement of anchor bolt drilling.
(d) Supplementary views as required showing the necessary dimensions and equipment.
(e) Identification plate list of parts having general application; complete list of material with name of all parts, part numbers, command or agency certificate of approval, drawing numbers or specification numbers, together with material for all parts included in standardized units. Total weights of each unit, with and without circuit breakers, shall be shown on drawing. All switches, circuit breakers, contactors, and resistors, shall be marked to show their current carrying capacity and voltage rating, where applicable.
(f) A one-line wiring diagram of main power circuits. Circuit breakers, disconnect links, main switches or other power circuit switching equipment shall be shown as required. Main and interconnecting buses may be shown.
(g) Detail drawings of appurtenances as required shall be shown or referenced. Where appurtenances are required to be of approved types for which test record drawings or certification data sheets are furnished the command or agency concerned, no working or corrected drawings of such appurtenances will be required; however, the shipbuilder or contractor shall reference each test record drawing or certification data sheet covering appurtenances, and the items in question shall conform completely to such drawings or certification data sheets, unless otherwise specified. If supplementary assembly, outline, or detail drawings of such items are necessary, in the judgement of the command or agency concerned, they shall be furnished by the switchboard manufacturer.
(h) Typical construction methods of the switchboard unit, showing details of framework, details of attaching front panels, details of joining structural members, location and types of welding, shall be shown or referenced.
(i) The drawings shall show piece numbers, identification plate numbers, and device numbers where necessary, and these shall agree with the corresponding numbers on the identification plate list and list of material. The list of material shall show the manufacturer's name and all sufficient information to identify readily the equipment in case replacement or purchase of new material is necessary.
(j) When switchboard drawings are not furnished, a wiring diagram of the unit shall be required.
(k) The drawings shall show the manufacturers recommended method for supporting incoming cables.

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3.24.2 Switchboard drawings. - Type I drawings of complete switchboards shall cover the information specified in categories A, B, C, D, F and G of MIL-D-1000, including the following:

(a) A dimensional front and plan view of the switchboard, showing assembly of all units comprising the switchboard, and overall dimensions of each section and type numbers of units as shown on Drawing 9000-S6201-74453, including a single line diagram showing bus sizes. This drawing shall show the weight and location of the center of gravity for each switchboard.
(b) Views as required showing necessary details not shown on drawings of the units, such as interconnection of buses between unit, dimensions and spacing of footing or mounting members, and other details.
(c) Identification plate list of parts having specific application; list of material for all specific and interconnecting parts not covered by master drawings, showing all required additional information, such as fuse or trip element data.
(d) Complete wiring diagram, showing feeder numbers, terminal blocks, circuit breaker, switch and fuse sizes, cable sizes and nameplate numbers. All electrical appurtenances shall be marked or indicated by piece numbers, to identify them readily with the corresponding part on the list of material. Where five or more single lines used to represent several leads parallel to each other on the plan, they shall be separated into groups (two or three to a group as convenient), to facilitate following with the eye, instead of having them all uniformly spaced.
(e) A single line diagram of metering, relaying, and control circuits. This diagram shall show the major bus circuits, the number and location of all instrument and control transformers, the instruments, relays and other equipment connected to each transformer, and the location of the instruments in relation to the various switchboards. The diagram shall be included among the drawings in the manual.
(f) Detail drawings of appurtenances not covered by drawings, as required. Where appurtenances are required to be of approved types for which test record drawings or certification data sheets are furnished the command or agency concerned, no working or corrected drawings of such appurtenances will be required; however, the switchboard manufacturer shall reference each test record drawing or certification data sheet covering appurtenances, and the items in question shall conform completely to such drawings or certification data sheets, unless otherwise noted. If supplementary assembly, outline, or detail drawings of such items are necessary, in the judgement of the Supervisor of Shipbuilding for the purpose of proper inspection or the identification of repair parts, they shall be furnished by the switchboard manufacturer.
(g) A typical construction drawing showing necessary details not covered by 3.24.1(h) or by other drawings.
(h) The drawings shall show piece numbers, identification plate numbers, and device numbers where necessary, and these shall agree with the corresponding numbers on the identification plate list and list of material. The list of material shall show the manufacturer's name and all sufficient information to identify readily the equipment in case replacement or purchase of new material is necessary.

3.25 Manuals. - The manuals shall conform to type I of MIL-M-15071 except as follows:

(a) Switchboard manuals shall, unless otherwise specified in the contract or order, contain all the information required in the following paragraphs and in the order listed: (Note. - The above applies where instructions for several switchboards are included in one book. If any of these switchboards are not furnished, the following chapter numbers shall be rearranged accordingly.)
Part 1 - Switchboards:
Chapter 1 - General information.
Chapter 2 - Switchgear group ship's service power.
Chapter 3 - Switchgear group-emergency power.
Chapter 4 - Miscellaneous (other switchboard sections if supplied.)
Part 2 - Air circuit breakers.
Part 3 - Miscellaneous switchboard devices.
Part 4 - Tools.
(b) Index. - The index shall be included at the beginning of each part, and shall be printed on a heavy paper spacer. The table of contents in the front of the book shall refer to this index.

Note. - Where list of illustrations and index are included in the device sections prepared by the manufacturer, reference shall be made to the list and index.

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3.25.1 Part 1 - Switchboards. - The chapter numbers below apply where instructions for several switchboards are included in one book.

Chapter 1 - General information.
(a) Introduction. - The introduction shall include a list of the various switchboards furnished, and description of their locations.
(b) Detailed description. - The detailed description shall include important mechanical details and features which have a bearing on the operation of the equipment.
(c) Principle of operation. - Discussion of principles of operation shall be based upon a simplified one-line diagram of the ship service system, emergency system, and other systems, and shall include a description of the sources of power, electrical arrangement of buses, purpose of the various switchboards, that is, ship service power, emergency power, and others. This chapter shall include a discussion of protective device arrangement and operation, selective tripping, such general features as synchronizing, ground detection, and indicating lamps, and such illustrations and sketches as are necessary to clarify the operation.

Chapter 2 - Switchgear group - ship service power - operating instructions. Include step by step procedure for starting generators, synchronizing and paralleling generators, dividing loads, securing generators, and use of ground lamps and indicating lamps.
Chapter 3 - Switchgear group - emergency power - operating instructions.
Chapter 4 - Miscellaneous - operating instructions for other switchboard sections, if supplied.

3.25.2 Part 2 - Air circuit breakers. - This part shall consist of a chapter for each frame size circuit breaker used on the switchboard, arranged as follows:

Chapter 1 - Type ACB - 250
Chapter 2A - Type ACB - 640R
Chapter 2B - Type ACB - 900RC
Chapter 2C - Type ACB - 901R
Chapter 2D - Type ACB - 902R
Chapter 3A - Type ACB - 1600R
Chapter 3B - Type ACB - 1600HR
Chapter 3C - Type ACB - 2601R
Chapter 4 - Type ACB - 3200HR
Chapter 5 - Type ACB - 4000HR
Chapter 6A - Type AQB - A-250
Chapter 6B - Type AQB - LF-250
Chapter 7A - Type AQB - A-100
Chapter 7B - Type AQB - LF-100
Chapter 7C - Type AQB - A-101
Chapter 8 - Type AQB - A-600
Chapter 9A - Type AQB - A-400
Chapter 9B - Type AQB - LF-400
Chapter 10 - Type AQB - A-50
Chapter 11 - Type ALB - 10 (3 pole)
Chapter 12 - Type ALB - 1 (single pole)
Chapter 13 - Type AQB - 800
Chapter 13A - Type AQB - A1601
Chapter 14 - Motor operated mechanism for AQB's.

These instructions shall be prepared by the circuit breaker manufacturer for insertion into the books.

3.25.3 Part 3 - Miscellaneous switchboard devices. - Switchboard devices shall be arranged by sections as follows, prepared by the manufacturer. The pages of each section shall be numbered consecutively. The sections shall be arranged in alphabetical order, but not necessarily lettered consecutively in the chapter.

Chapter 1 - Instruments, instrument transformers and instrument accessories.
Section A - D.C. ammeters and voltmeters.
Section B - A.C. ammeters and voltmeters.
Section C - Wattmeter.
Section E - Frequency meter.

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Section F - Synchroscope.
Section G - Temperature meter equipment.
Section H - Instrument transformers.
Section J - Current transformer protective device.
Section K - Shunts.
Chapter 2 - Switches and devices. - A general write-up on control switches and indicator lights, test blocks and plugs, and bus disconnecting switches, giving descriptive data, illustrations, rating, without specific reference to any one particular switchboard.
Chapter 3 - Relays.
Section A - Protective relays.
Section B - Voltage control relays.
Section C - Current control relays.
Section D - Auxiliary relays.
Section E - Bus transfer (where not covered above.)
Chapter 4 - Rheostats and miscellaneous.
Part 4 - Special tools. - A list of special tools with their application should be included.

3.25.4 Maintenance. - Maintenance of equipment shall be included in the section for each particular piece of equipment. These sections shall show complete information for enabling laymen to effect efficient maintenance of equipment, including service inspection, lubrication, and routine maintenance.

3.25.4.1 Drawings. - Drawings and illustrations shall include the following:

(a) A front view arrangement plan of each switchboard showing in simplified fashion, arrangement, part identification and nameplate number of each device on front of switchboard. The nameplate identification shall agree with the switchboard plan. (See part 1, chapters 2 to 4 inclusive, herein.) This plan may be a reduction of the switchboard front view plan.
(b) A simplified one-line diagram showing only arrangement of primary connections for generators, bus ties, buses, typical feeders, and shore connections. Each circuit breaker or device shown on this diagram shall have nameplate identification related to (a) above. No control or metering circuits shall be shown except for very simple systems where the inclusion of control and metering circuits will not complicate the drawing. For simple systems this diagram may be a reduction of the switchboard complete wiring diagram.
(c) Metering and control diagram: This diagram may be reduction of the switchboard complete wiring diagram. When a specific drawing diagram is furnished, it shall show the major bus circuits, the number and location of all instruments and control transformers, the instruments, relays and other equipment connected to each transformer, and the location of instruments in relation to the various switchboards.
(d) A diagram showing typical circuits, such as synchronizing, and ground lamps, to be separately illustrated and discussed in the text under part 1, chapter 1.

3.26 Repair parts. - Repair parts required for the maintenance of the equipment shall be supplied as onboard repair parts. Onboard repair parts list and cataloging information shall be furnished in accordance with MIL-P-15137. Onboard repair parts shall be furnished in accordance with table V.

Table V - Onboard repair parts.

Item	To determine the quantity of onboard repair parts to be furnished for ship or on an order, multiply the quantity of identical installed replaceable parts times the assigned multiplying factor and round the product to the nearest whole number. For numbers less than one, furnish one item.
	Multiplying factor
Fuseholder, each size and type	.05
Fuses, each size and type (except fuses for AQBLF circuit breaker, see MIL-C-17361)	.50

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Table V - Onboard repair parts (Cont'd.).

Item	To determine the quantity of onboard repair parts to be furnished for ship or on an order, multiply the quantity of identical installed replaceable parts times the assigned multiplying factor and round the product to the nearest whole number. For numbers less than one, furnish one item.	
	Multiplying factor	
Lens, indicator light, each size and type	.10	
Light, indicator (where transformer or resistor are integral)	.05	
Lamps, incandescent, each size and type	2	
Relay coils, each size and type	.10	
Relay contacts, each size and type	.10	
Relay springs, each size and type	.10	
Resistor, each size and type	.10	
Rotary switch contacts, stationary and moveable	.5	
Rotary switch springs, each size and type	.10	
Transformer protective device, each size and type	1	
Special tools (one set furnished for each ships service and emergency switchboard)		
ACB and AQB/AQBLF circuit breaker repair parts shall conform to MIL-C-17587 and MIL-C-17361 respectively		
Bus transfer equipment repair parts shall conform to MIL-S-17773		

3.27 Tools. - Ordinarily no tools are required to be furnished with switchboards. Where the construction of the switchboards or any of the appurtenances installed thereon is such as to require the use of certain special tools for their proper service maintenance, one complete set of such special tools shall be furnished for each ship's service and emergency switchboard installed.

3.28 The Government will furnish material as specified (see 6.1).

3.29 Qualification under reference specifications. - When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which have been approved for listing on the applicable Qualified Products List on the date of invitation for bids or date of initiation of purchase action shall be utilized. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The supplier shall have a quality control system that complies with the requirements of MIL-Q-9858.

4.2 General inspection. - The equipment shall be given such tests and inspection as will verify and assure conformance with this specification. All materials and components, both electrical and structural, shall be examined and tested in accordance with the applicable specifications either at the various sources or at the place of assembly.

4.3 Inspection. -

4.3.1 Examination and tests of assembly. - Switchboards shall be completely assembled and shall be examined to insure compliance with this specification, using for guidance the classification of defects contained in table VI. The switchboards shall be tested to discover any incorrect connections, open circuits, short circuits, grounded circuits, and incorrect operation of the various apparatus. The insulation resistance shall be measured and be not less than the requirements of the particular equipment specification or 10 megohms at 25° C, whichever is applicable.

Table VI - Classification of defects.

Categories	Defects
Critical:	
1	Enclosures, panels, compartmentation not provided to prevent injury from electric shock; rear enclosure perforations, ventilating louvers exceed size limit.
2	Grab rails and guard rails not located or mounted as specified to insure safe operation.
Major:	
101	Type, number or section arrangement nonconforming.
102	Section unit not complete, appurtenances or component parts missing.
103	Evidence of use of unauthorized material.
104	Parts broken, cracked, distorted or chipped.
105	Control benchboard (if required) dimensions, arrangement of appurtenances nonconforming.
106	Structure not in accordance with approved drawing; not framed, bolted, welded or dimensioned as required; gusset plates nonconforming.
107	Framework bracing not as specified; bolt size less than minimum; unauthorized use of rivets.
108	Standard units or special sections not bolted to foundation channel in accordance with approved drawing; number and size of bolts nonconforming.
109	Side sheets less than specified thickness; welding to frame nonconforming, bus bar cutouts not located in accordance with applicable drawing.
110	Compartmentation not as specified; circuit breakers not isolated; insulation distance not as specified.
111	Ventilation not adequate; openings not size for circuit breaker load; openings not screened.
112	For submarine applications, enclosure does not meet spraytight requirements; construction of equipment impairs operator's view of instruments or restricts its operation.
113	Locking devices not acceptable type; not provided where required.
114	Panel thickness, size, forming, stiffening or location nonconforming.
115	Panel megger holes not provided type AQB and NQB circuit breaker panels; not hinged as specified, when required.
116	Equipment brackets, bolting or arrangement nonconforming; switches not mounted in specified plane; or phases not disposed as required.

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Table VI - Classification of defects (Cont'd.).

Categories	Defects
Major:	
117	Casualty power connection, if required, not accessible; mounting nonconforming; tool rack not outside enclosure, casualty power circuit breaker not as specified.
118	Voltage regulator mounting nonconforming.
119	Bus bars not size, cross-section required by ampere rating; not silver surfaced, formed as specified; ends not neatly furnished; bolt hole contact areas not flat; bolt size and number not as specified.
120	Cable connectors not specified type and size for required load and use.
121	Bus bar assemblies not painted with specified coating; not applied as required, or in lieu of painting, bus bar not marked with specified warning.
122	Evidence that bus bar supports not spaced in accordance with short circuit current available.
123	Evidence that bus bar capacity nonconforming.
124	Cables connectors not specified type; solder, if authorized, not as specified.
125	Means not provided for connecting cables from other units or sections; cable connection arrangement restricts access.
126	Instrument and control wiring nonconforming; not connected as specified; not grouped and strapped as specified.
127	Terminal boards not provided where required.
128	Synchronizing circuits, where required, not connected in specified manner; synchronizing control equipment not as specified.
129	Indicator lights, space heater circuits, transformers, voltage regulators and trip circuits for ACB circuit breakers not connected or arranged in accordance with applicable specifications or approved drawing.
130	Wiring not marked or tagged.
131	Circuit breakers not type specified; location and mounting nonconforming; draw-out mechanism (for type ACB only) not operable as specified.
132	Switches not in accordance with specification; instrument control switches not type or not marked as specified; handle not required shape.
133	Disconnect devices not specified type; not mounted as required; tools, if required, not located properly.
134	Fuses not in accordance with specification; location identifications and mounting nonconforming.
135	Fuse holders, where required, not enclosed or not in accordance with approved drawing.
136	Terminal boards exceed allowable length.
137	Instruments not in accordance with specification; case, scale and mounting nonconforming.
138	Transformers not in accordance with applicable specifications; not provided; not connected as specified; not protected where required.
139	Relays not as specified; not approved type.
140	Indicator lights nonconforming; not color required by code; resistors not as specified; lamps not provided.
141	Rear and top enclosures not secured firmly to frame; will not clear live bus work if buckled; top flanges do not prevent water from seeping into the enclosure; panel fastening not as specified, method employed for making thumbscrews captive inadequate.
142	Top enclosure shape, form and flanges nonconforming; corners not welded watertight.
143	Buses not marked to indicate polarity or phase.
144	Appurtenances where so required by applicable specification not approved for Qualified Products Lists.
Minor:	
201	Painting procedure specified not followed; primer not applied as specified; enamel not specified type.
202	Identification plates not specified shape, size, thickness, location, letter and fastening.
203	Special tools, if required, not furnished.

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4.3.2 Vibration test. - The vibration test shall be conducted in accordance with the type I requirements of MIL-STD-167. One unit from each group specified in table VII shall be tested, as applicable. If the unit satisfactorily passes the vibration test, the other units within that group need not be tested. Any major redesign of the tested unit which might alter the ability of the units to withstand this test shall require a retest to determine compliance with these requirements. Separate tests shall be conducted on steel and aluminum switchgear units.

Table VII - Shock and vibration test grouping of switchgear units.

Group number					
1	2	3	4	5	6
G1	G2 (ACB)	G3	G5	D80	D85
G2 (AQB)	G11 (ACB)	B21	G7	D81	D86
G10	D50	B22	B25	D82	D87
G11 (AQB)	D60	D51	B26	D83	D95
E40	D61	D52	D56	D88	
E40A	D62	D55		D89	
	D63	D70		D90	
	D64	D71		D91	
	B20	D72		D92	
	E41	D73		D93	
		D74		D96	
		E42		D97	

4.3.3 Dielectric strength. - The equipment shall withstand for a period of 1 minute a dielectric test voltage of twice the rated voltage of the equipment plus 1,000 volts r.m.s. value. The frequency of the test voltage shall be not less than 60 cycles. It shall approximate a true sine wave. The source of the test potential shall have a rating of at least 1 kw.

4.3.4 High impact shock test. - All equipment required by this specification to be installed in the switchgear units shall have passed the high-impact shock test, as applicable, in accordance with the various equipment specifications. Such equipment does not have to be subjected to shock testing with the switchgear units. Suitable dummy weights may be substituted for this equipment prior to the shock test. Appurtenances which have not been previously shock tested individually, will be accepted when installed within the switchgear unit and found to pass the shock test of that unit. High-impact shock tests in accordance with MIL-S-901, shall be conducted on one unit of each group in table VII. If the unit satisfactorily passes the high impact shock test, the other units within that group need not be tested. Separate tests shall be conducted on steel and aluminum switchgear units. The test shall include mounting on a 30 degree incline and be conducted with the unit de-energized. The features of the test shall be as follows:

(a) The required type of shock test - type A.
(b) The weight designation of shock test - medium weight.
(c) Required class of equipment - class I - deck mounted.
(d) Required grade of equipment - grade A.

4.3.4.1 Definition of failure to perform principal function. - Sufficient electrical tests and visual examinations shall be made to determine that all components installed within the switchgear unit are capable of performing their intended function. The framework members shall show no evidence of fractures and any resulting distortion shall not be of sufficient magnitude to impair the operation of the equipment contained within. The enclosure shall not distort to an extent which might result in contact with any electrically conducting part within the unit. The bus work shall not be mechanically damaged nor the bus clearance distances reduced below the minimum requirements.

4.3.5 Protection of enclosure from waterspray for submarine applications (see 6.1). -

4.3.5.1 Waterspray test. - The switchgear shall continue to operate when subjected to the waterspray test requirements of MIL-STD-108 for spraytight enclosure with the following modifications:

(a) Direct the stream of fresh water at representative covers, handles, seams, plates, etc. on the front, back, sides, and top of the enclosures at all angles and all positions for 1 hour. The top of the switchgear shall be tilted 30 degrees towards the rear when testing the front surface.

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(b) Measurements shall be made of the amount of water which entered the switchboard and its location noted.

(c) Resistance to ground readings of each electrical circuit shall be taken. When the manufacturer has developed an approved design for making switchgear enclosures spraytight, the waterspray test shall be conducted on one representative unit of this construction. If the methods employed by the manufacturer to achieve spraytight construction in an approved design are modified, a retest shall be conducted to re-establish that the spraytight requirements are met.

4.3.5.2 Heat test (see 6.1). - Heat run tests shall be conducted to ensure that the maximum operating temperature limits of the switchgear components (including semiconductors, insulation, and current carrying connections) are not exceeded, based on the specified outside ambient temperature. The test shall be conducted on one complete switchgear group of an approved spraytight construction, under conditions simulating a shipboard installation. The switchgear groups shall be operated at full load with all necessary cable connections to the various circuit breakers and with the bottom sealed off. Sufficient temperature detectors (thermocouples or other approved methods) shall be installed to determine the maximum temperature of the switchboard components, as well as the ambient temperatures of the various compartments within the units. The test shall be of sufficient duration to allow stabilization of the measured temperatures. Approval of this test will be extended to subsequent units manufactured which use the same basic methods to achieve spraytight construction and ventilation.

4.4 Inspection of preparation for delivery. - The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment and accessories. -

5.1.1.1 Preservation and packaging. - Equipment and accessories shall be preserved and packaged to provide adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early equipment installation and may conform to the suppliers' commercial practice when such meets these requirements.

5.1.1.2 Packing. - Equipment and accessories shall be packed in a manner which will be acceptable by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers' commercial practice when such meets these requirements.

5.1.1.3 Marking. - In addition to any special marking required by the contract or order, shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The marking information shall include Federal Stock number or manufacturer's part number; nomenclature; contract or order number; contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged by level A, packed by level A or B (see 6.1) and marked in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking. - Equipment and accessories, replacement devices, stock repair parts and technical publications shall be preserved and packaged by levels A or C; packed by level A or B as specified and marked in accordance with MIL-E-17555).

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, number, and arrangement of units comprising the switchgear sections (see 3.3 to 3.7, inclusive).
(c) Sections comprising a switchgear group (see 3.3 to 3.7, inclusive).
(d) Power, current, and voltage rating of generators controlled and quantity of casualty power equipment (see 3.3.1, 3.3.2, and 3.8).
(e) Whether vertical control switchboard is required (see 3.3.5).
(f) Feeders, breakers, and switches (see 3.3.6, 3.3.7, 3.3.8, 3.5, 3.6, and 3.7.3).
(g) Requirement for type AQB circuit breaker on emergency switchgear for diesel cooling water booster pump (see 3.4.1).
(h) Requirement for electrically operated emergency bus unloader switch (see 3.4.1 and 3.4.2).
(i) Rating for shore power connection circuit breaker (see 3.3.3(h), 3.3.4.2(a) and 3.7.2.2).
(j) Material to be used in construction of switchgear structure and switchgear buses (see 3.8.11 and 3.13).
(k) Buses to be provided in each switchgear section (see 3.13).
(l) Cable lugs (see 3.13.15).
(m) Location of cable entrance (see 3.13.17).
(n) Circuit breaker frame size (see 3.15).
(o) Circuit breaker trip element ratings and tripping characteristics (see 3.15).
(p) Spraytight enclosure and heat run test for submarine applications (see 3.20.4 and 4.3.5).
(q) The following items are not a part of any individual standard unit. They will be supplied by the switchboard manufacturer only when complete switchgear sections are ordered. Otherwise they will be supplied by the command or agency assembling the standard units into switchgear sections:

    (1) Foundation channels.
    (2) Top sheet and drip shield.
    (3) Main bus bars.
    (4) Main bus supports, main bus spreaders and bus opening cover on end units.
    (5) Cable and wiring between units.
    (6) End trims.

(r) Whether delivery of drawing originals is required and the type of reproductions (see 3.24).
(s) List of Government furnished material (see 3.28).
(t) Preservation and packaging, packing, and marking if other than as specified in 5.1 (see 5.2).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 6110-N115 Sh)

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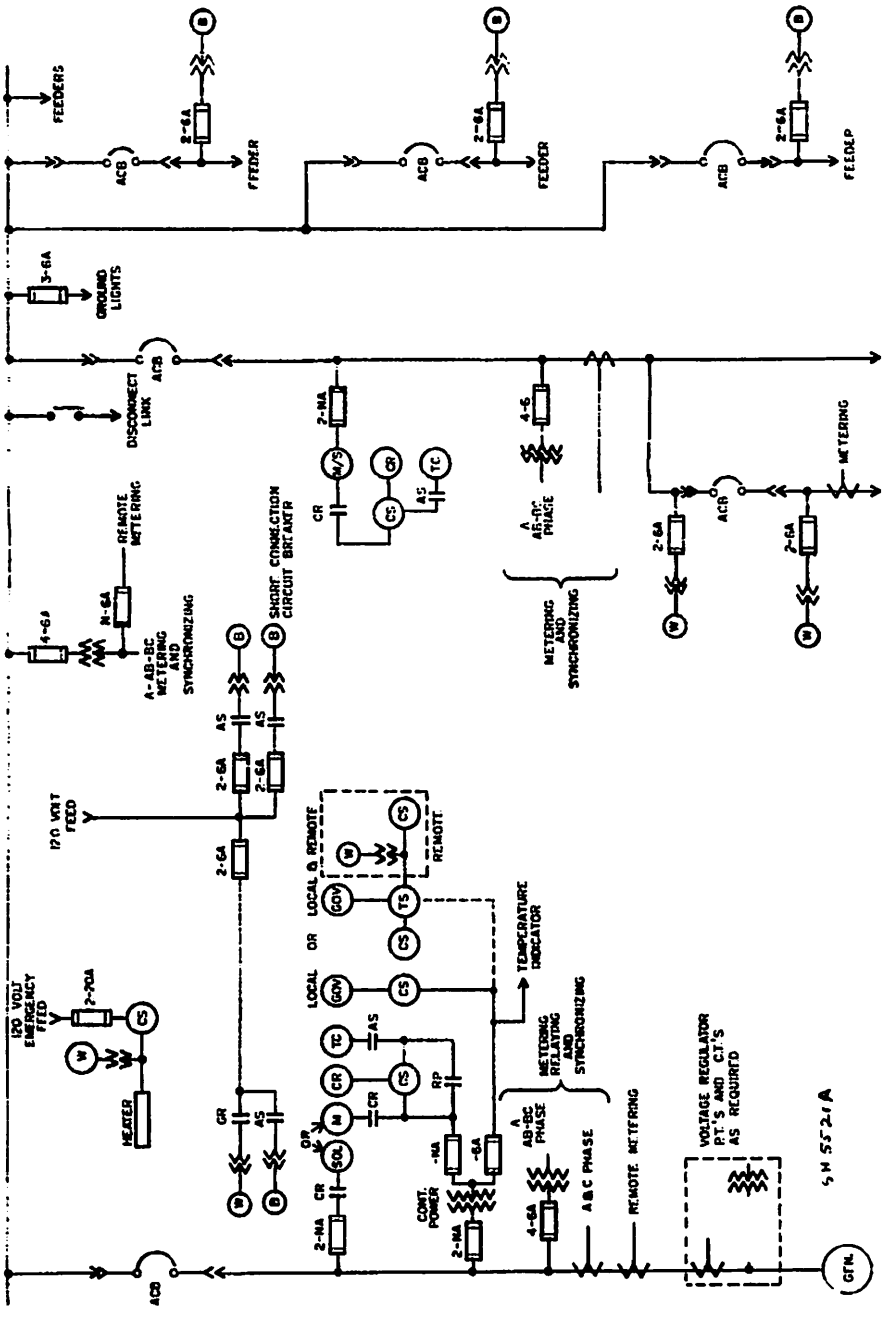


Figure 1 - Typical one line control diagram for ship's service a.c. power switchgear group having a control benchboard (see figure 2 for legend).

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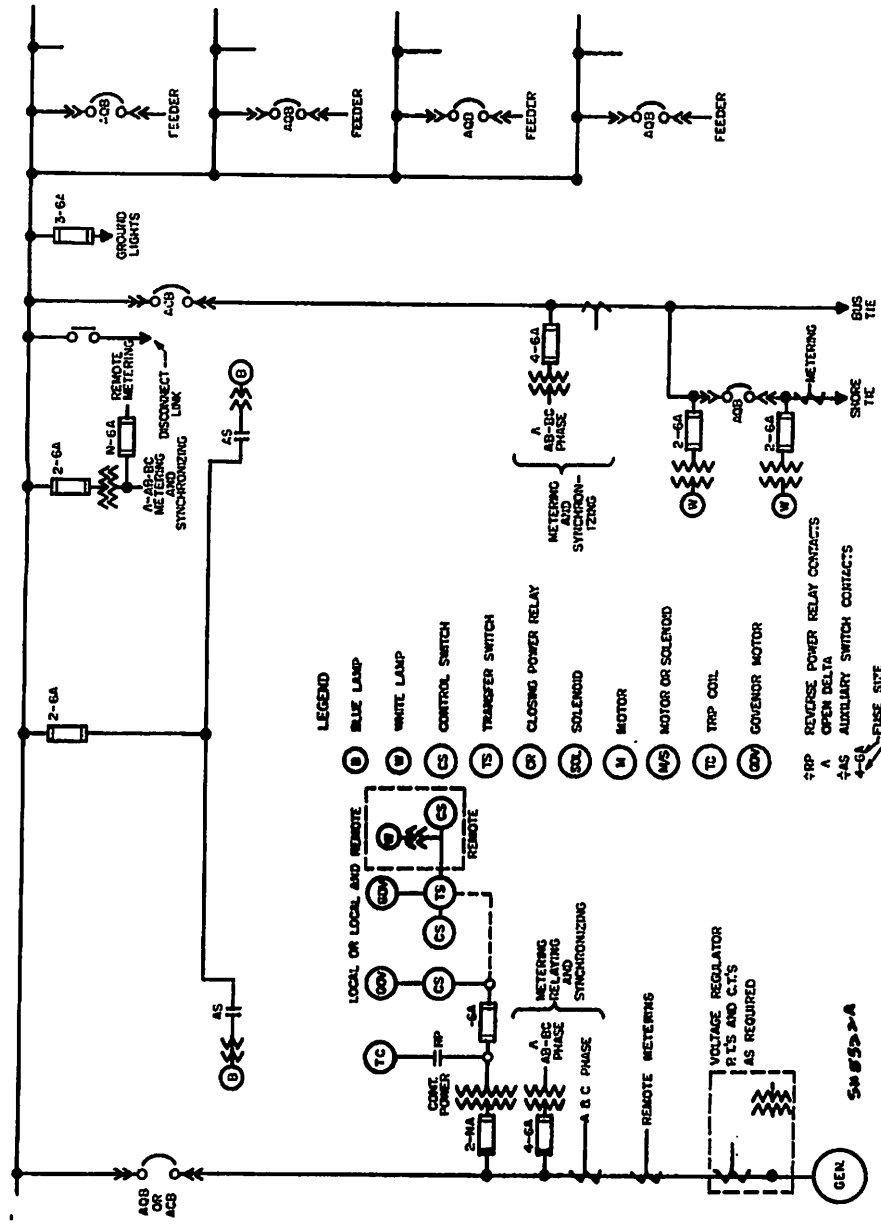


Figure 2 - Typical one line control diagram for ship's service a.c. power switchgear group not having a control benchboard.

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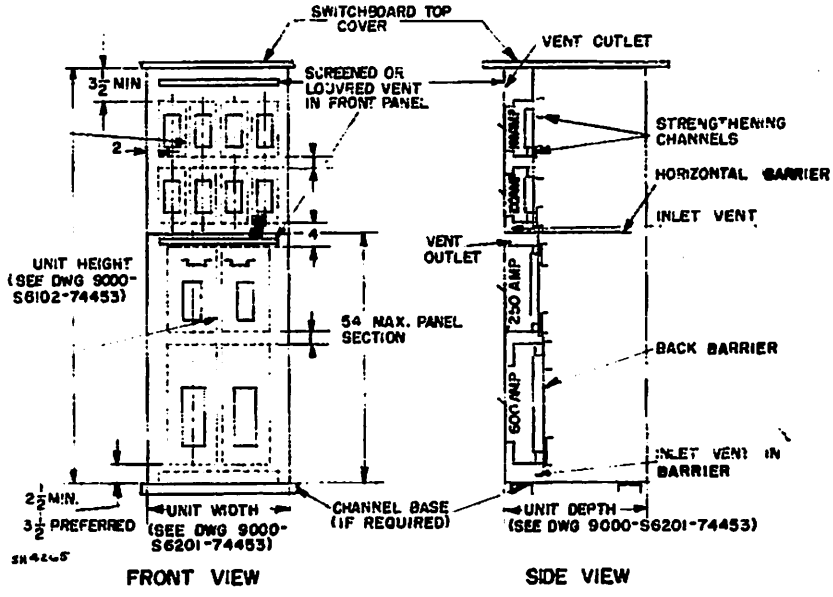


Figure 3 - Distribution unit type AQB circuit breakers (dimensions and spacings).

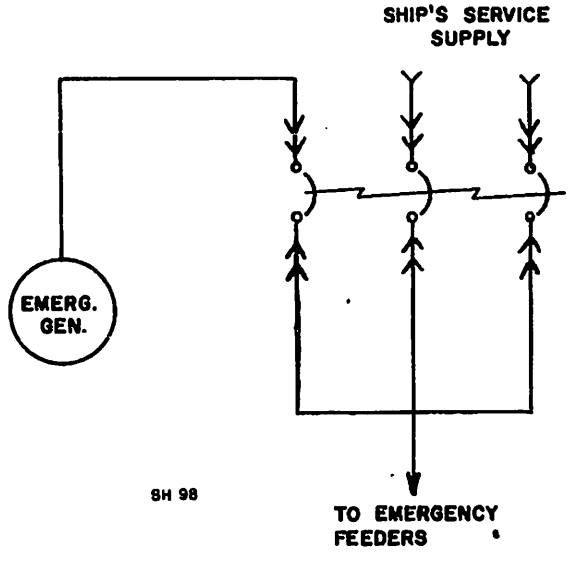


Figure 4 - Three-way bus transfer type ABT-A3(ES), circuit breaker type.

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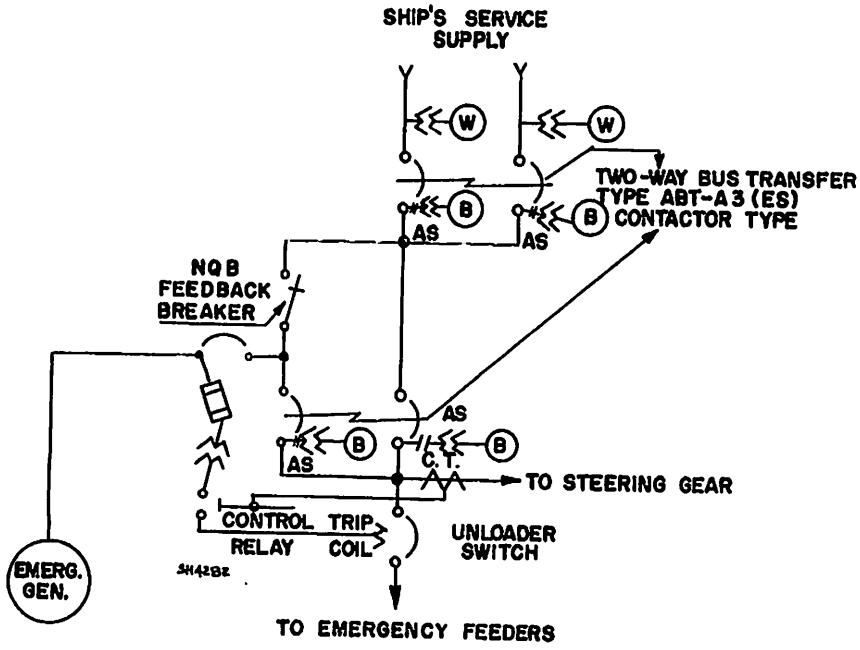


Figure 5 - One line diagram of emergency power system.

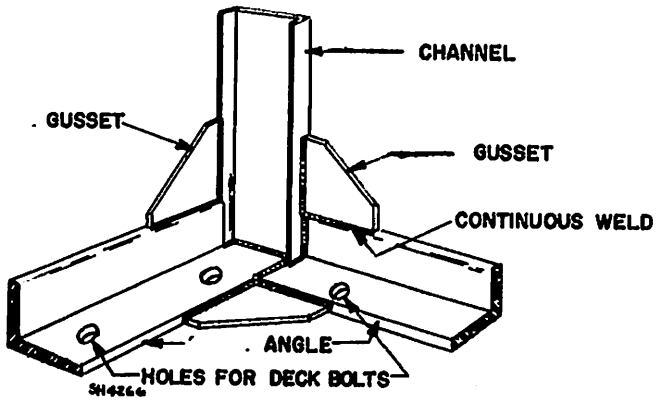
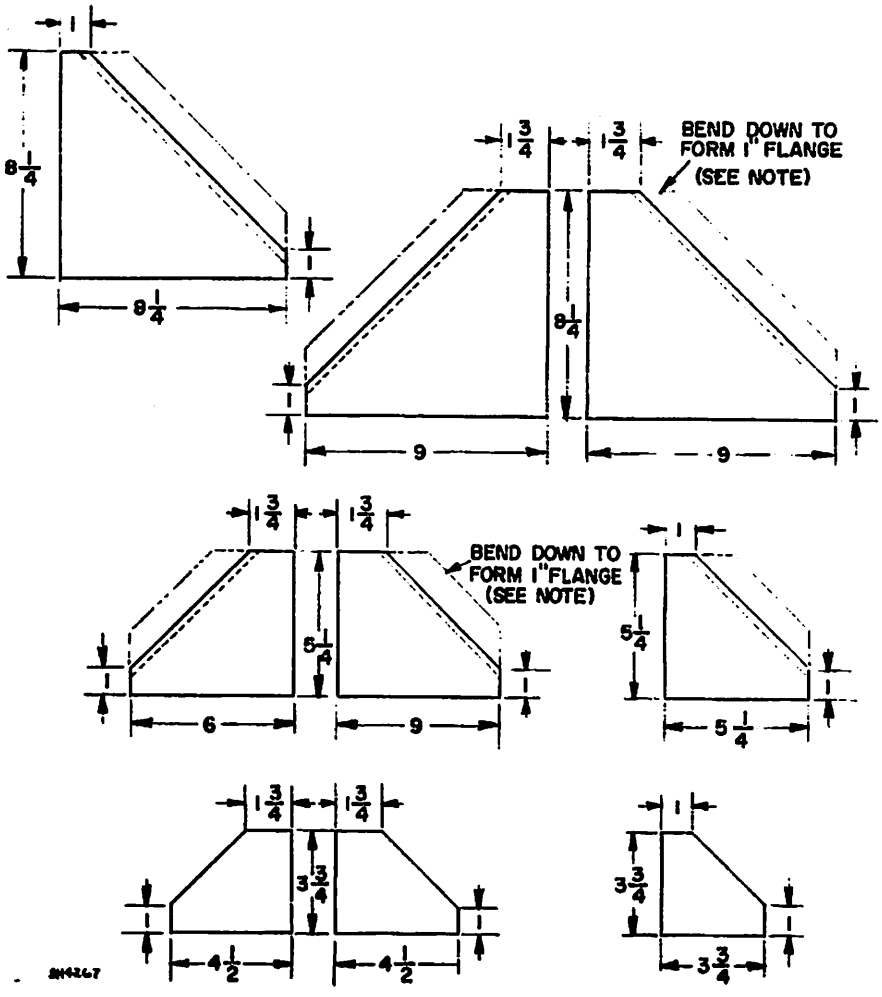


Figure 6 - Typical bottom corner construction.

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NOTE: Flanges shall be used when thickness of gusset plates is less than 3/16 inch. Gusset plates shall not be less than 1/8 inch thick.

Figure 7 - Typical gusset plates.

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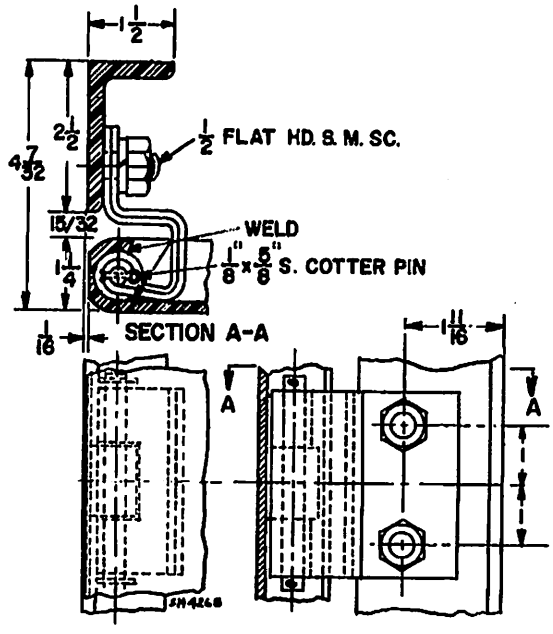


Figure 8 - Typical panel hinge assembly.

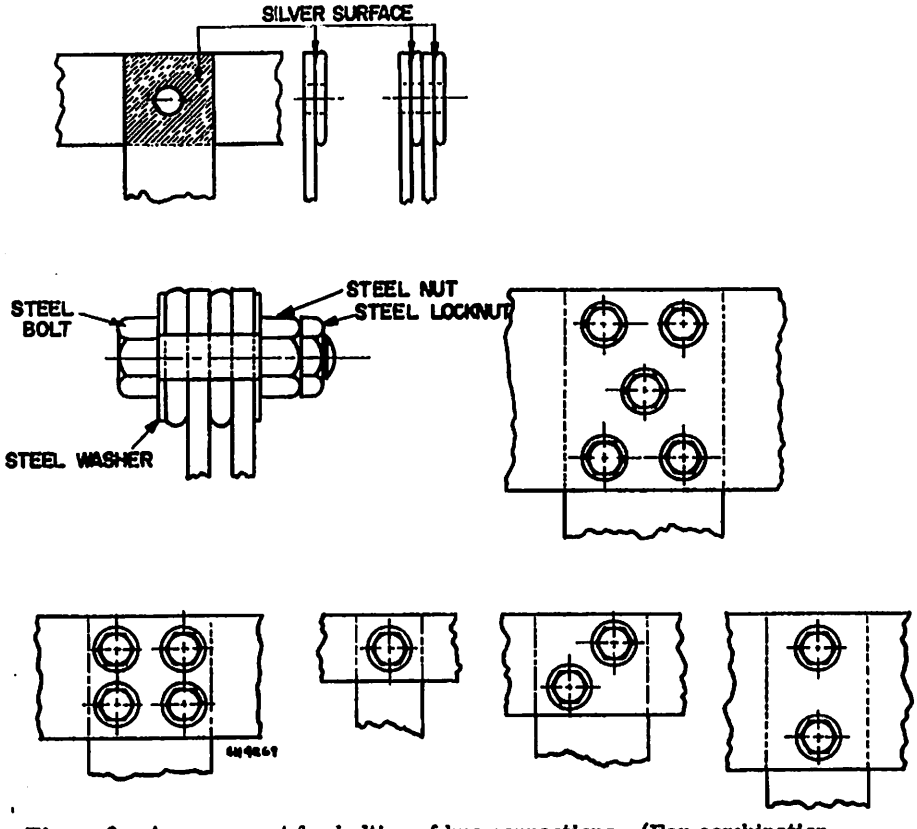


Figure 9 - Arrangement for bolting of bus connections. (For combination of bolt and bus sizes see table III.)

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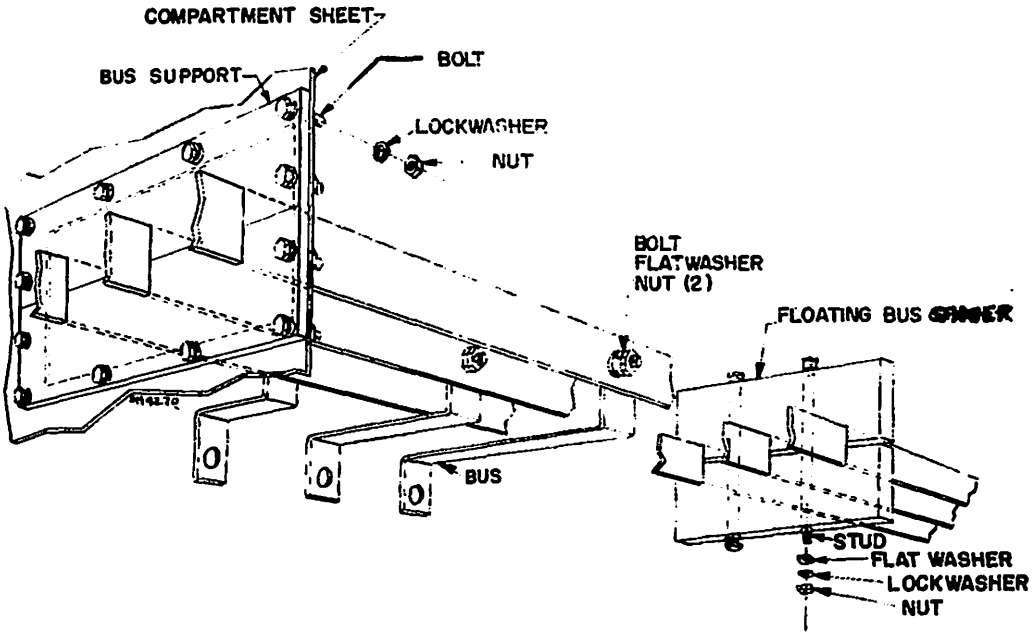
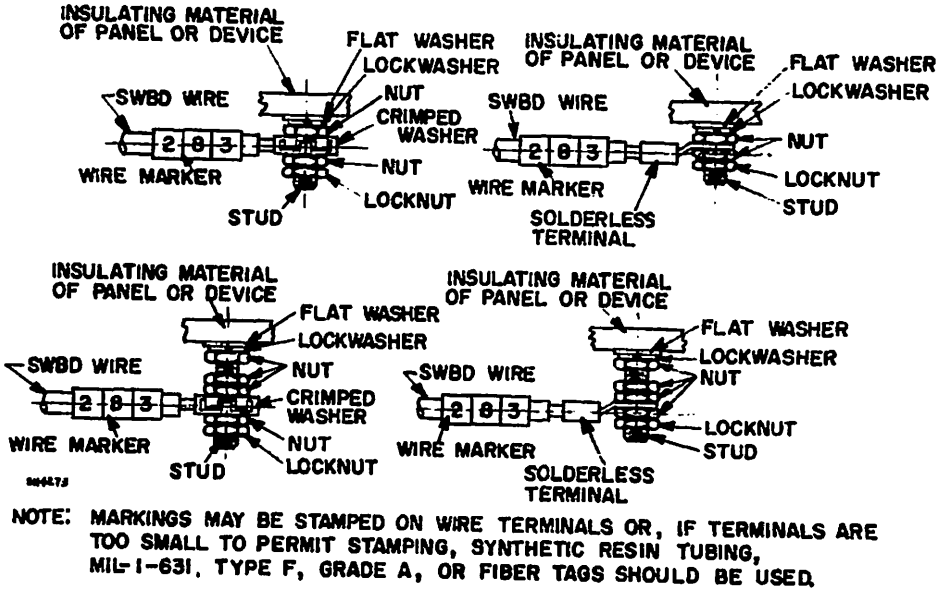


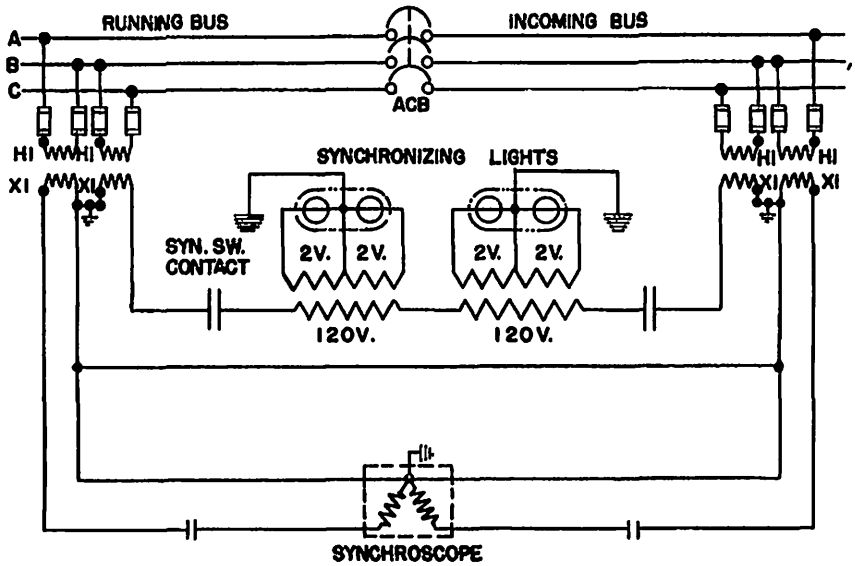
Figure 10 - Typical bus bar mountings.



NOTE: MARKINGS MAY BE STAMPED ON WIRE TERMINALS OR, IF TERMINALS ARE TOO SMALL TO PERMIT STAMPING, SYNTHETIC RESIN TUBING, MIL-I-631, TYPE F, GRADE A, OR FIBER TAGS SHOULD BE USED.

Figure 11 - Cable marking and detail of connections.

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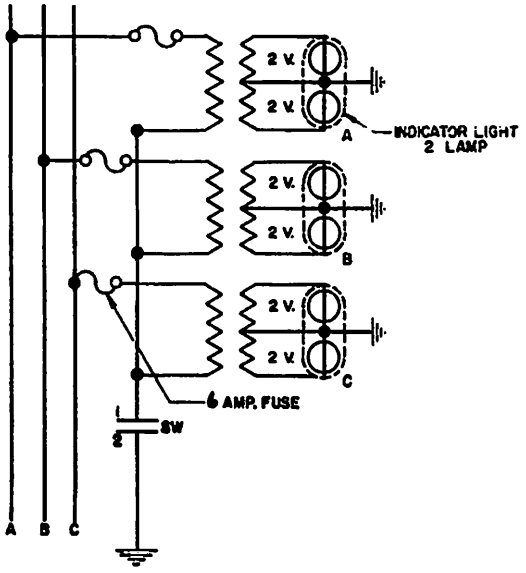


NOTE: THIS DIAGRAM IS TYPICAL ONLY, REPRESENTING A REFERENCE BUS AND A SECOND BUS TO BE SYNCHRONIZED WITH IT. WHERE PROVISION MUST BE MADE FOR A THIRD BUS TO BE SYNCHRONIZED WITH EITHER OF THESE BUSES, ADDITIONAL CONTACTS IN THE SYNCHRONIZING TRANSFER SWITCH SHALL BE PROVIDED AS REQUIRED.

SH 99

Figure 12 - Synchronizing connections.

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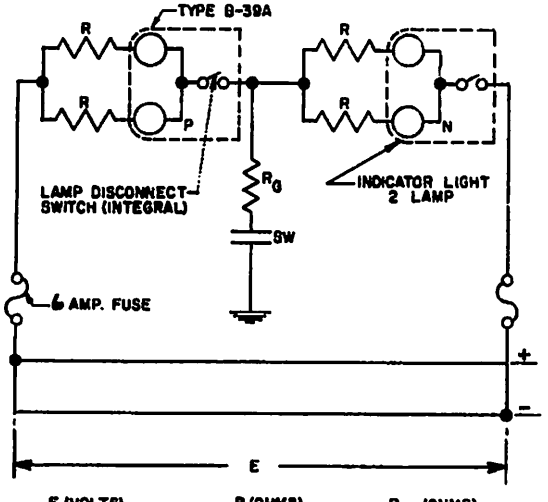


NORMAL–ALL LIGHTS DIM
PHASE A GRD–"A" LIGHT DARK
"B" LIGHT BRIGHT
"C" LIGHT BRIGHT
PHASE B GRD–"B" LIGHT DARK
"A" LIGHT BRIGHT
"C" LIGHT BRIGHT
PHASE C GRD–"C" LIGHT DARK
"A" LIGHT BRIGHT
"B" LIGHT BRIGHT

CONTACTS		POSITIONS	
		TEST	NORMAL
1 2	1	X	
	2	X	

SPRING RETURN

A.C. GROUND LIGHTS
3 PHASE



NORMAL–BOTH LIGHTS DIM
POS. GRD –"P" LIGHT DARK
"N" LIGHT BRIGHT
NEG. GRD –"N" LIGHT DARK
"P" LIGHT BRIGHT

CONTACTS		POSITIONS	
		TEST	NORMAL
1 2	1	X	
	2	X	

SPRING RETURN

SH 1008

E (VOLTS)	R (OHMS)	R_G (OHMS)
120	1100	630
240	2800	1400

D.C. GROUND LIGHTS
2 WIRE

Figure 13 - Ground light schemes.

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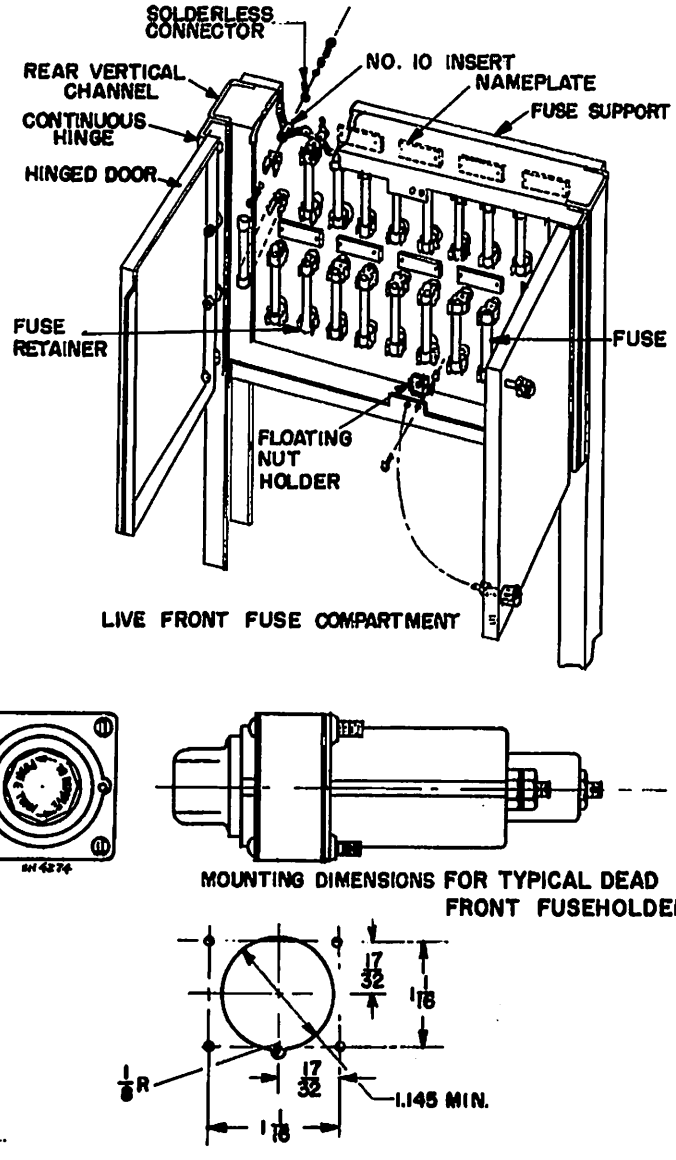


Figure 14 - Fuse mounting.

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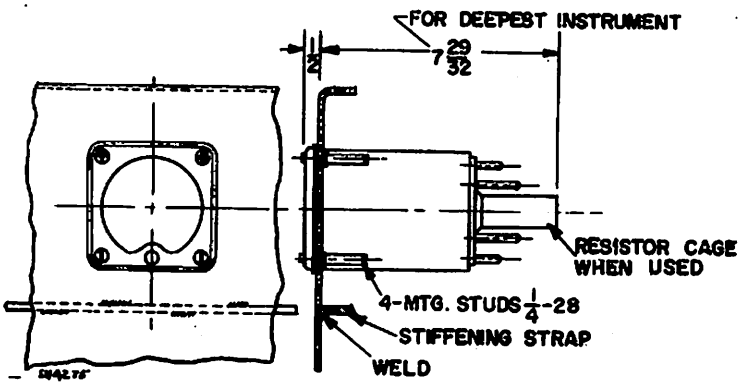


Figure 15 - Method of mounting switchboard instruments.

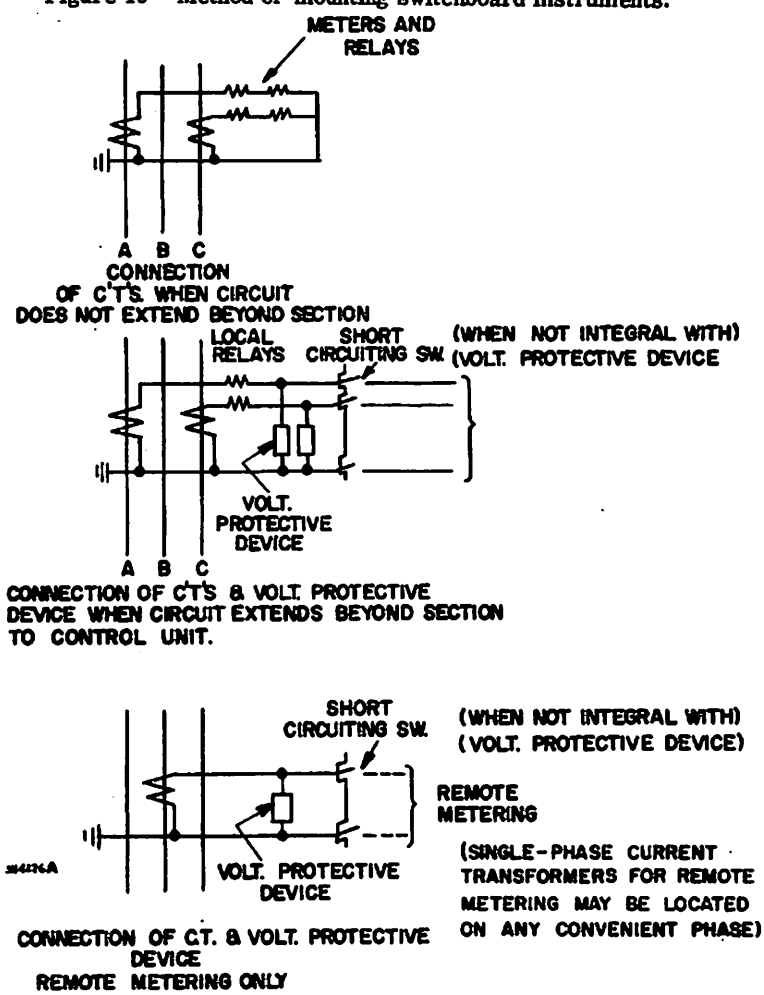


Figure 16 - Connections of current transformers and voltage protective devices.

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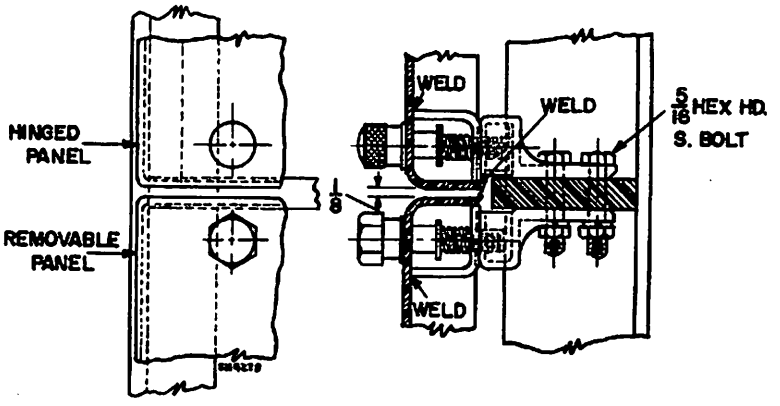


Figure 17 - Panel latch bolt for hinged and removable panel.

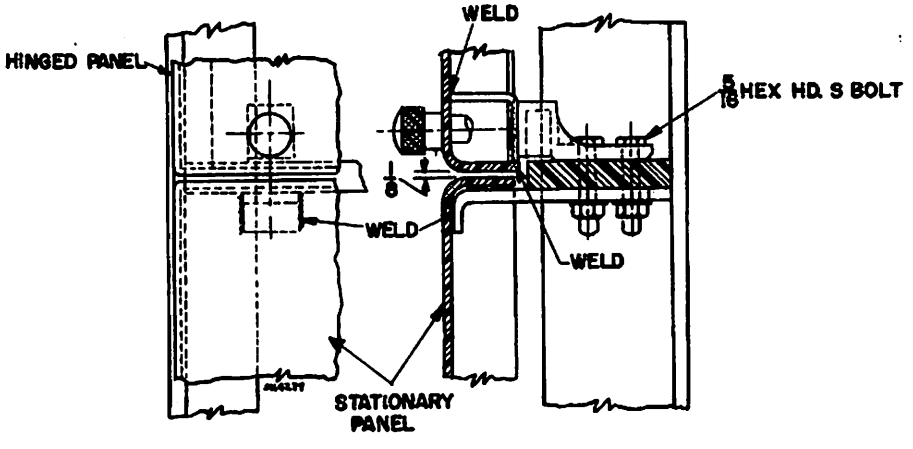


Figure 18 - Panel latch bolt for hinged panel and method of securing stationary panel.

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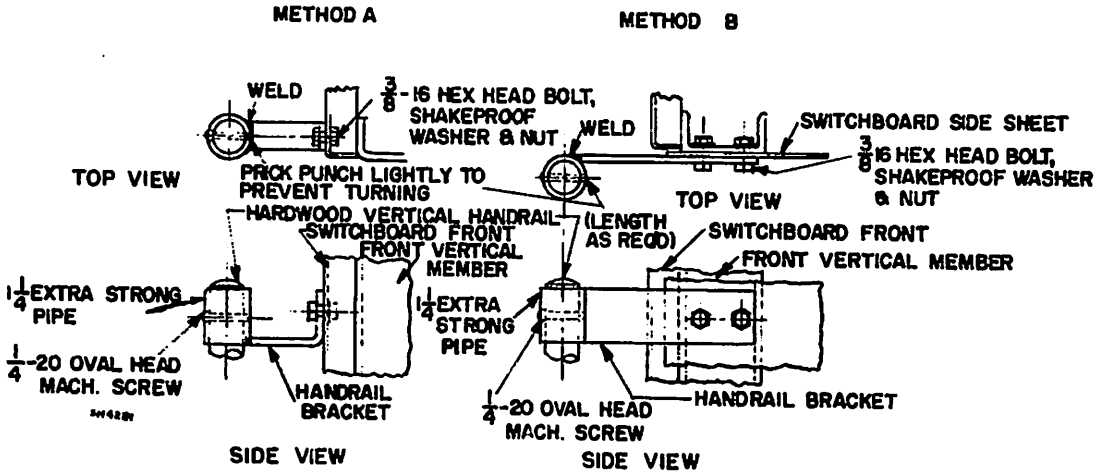


Figure 19 - Typical vertical guard rail mounting.

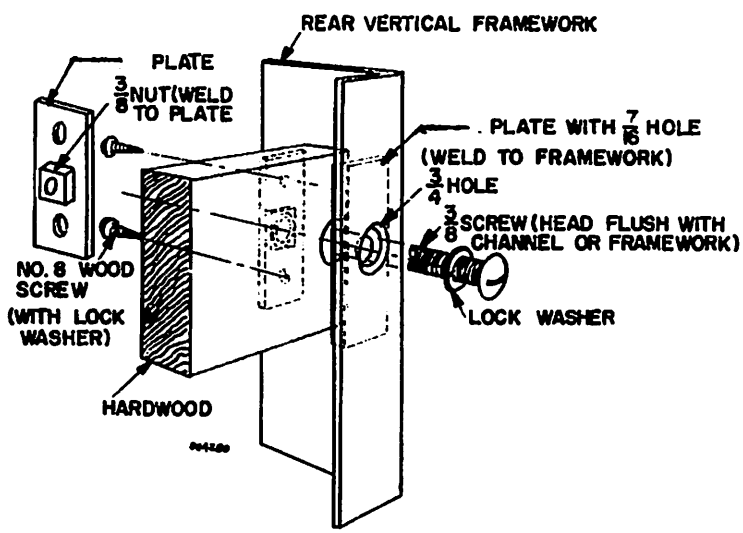


Figure 20 - Typical rear guard rail mounting.

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